

Materials Design and Assessment of Redox-Mediated Flow Cell Systems for Enhanced Energy Storage and Conversion

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The transition toward sustainable energy systems necessitates innovations that overcome the limitations of conventional electrochemical systems. Redox-mediated flow cell systems emerge as a transformative paradigm by decoupling energy storage, conversion, and chemical processes from traditional electrode-bound reactions. These systems employ soluble redox mediators to shuttle electrons between electrodes and spatially separated reactive phases (solid, liquid, or gas), thereby enabling unprecedented operational flexibility and scalability. This standpoint underscores the adaptability of redox-mediated electrified systems across a range of applications, encompassing high-energy-density redox targeting-based flow batteries, fuel cells, electrified CO₂ capture, sustainable chemical synthesis, waste recycling, etc. The rational design of redox-active materials is central to their success, with precise alignment of redox potentials, enhanced electron-transfer kinetics, and robust stability underpinning performance. The challenges of new materials development, system durability, and cost-effectiveness can be addressed through advances in experimental measurement, computational modeling, operando characterization, and interdisciplinary collaboration. Moving forward, the integration of redox-mediated technologies with renewable energy systems and industrial processes is predicted to transform energy and chemical landscapes. The integration of laboratory innovations with real-world deployment facilitates a pathway to decarbonization, resource efficiency, and the circular economy. This perspective emphasizes the pivotal functions of redox-mediated architectures in fostering a robust, electrified future, where the convergence of energy storage, environmental stewardship, and sustainable chemical production is pivotal in addressing global challenges.

1. Introduction of Redox-Mediated Flow Cells

The growing global energy demand and the urgent transition toward sustainable technologies have spurred a wave of innovation in electrochemical systems for energy storage and conversion.^[1–3] However, conventional electrochemical systems are hindered by structural and operational limitations inherent to their designs, particularly for large-scale applications. In typical electrochemical setups, energy conversion and storage occur at the electrode-electrolyte (E-E) interface, where reaction kinetics and capacity are constrained by the physical dimensions of the electrode compartments.^[4,5] As shown in **Figure 1a**, these confined geometries inherently limit the available active surface area, restrict mass transport, and cap the amount of active material that can participate in the reactions. Furthermore, electrode materials or electrocatalysts are often required to be electrically conducting or semiconducting, narrowing the pool of viable candidates. The interface itself becomes increasingly complex in systems involving multiphase reactions, such as gas evolution in water electrolysis or oxygen reduction in fuel cells, where three-phase contact (solid-liquid-gas) and dynamic interfacial restructuring further complicate charge transfer and product management.^[6–8] Collectively, these limitations curtail the

scalability, energy density, and operational flexibility of traditional electrochemical systems.

Redox-mediated flow cells offer a transformative alternative to conventional electrochemical architectures by shifting the targeted chemical reactions away from the electrodes.^[9–12] Instead of relying on interfacial electron transfer at the E-E interface, these systems employ soluble redox mediators (RMs) that shuttle electrons between the electrode and a spatially separated reactive material, which may be present in a different physical phase: solid, liquid, or gas. As shown in **Figure 1b**, this fundamentally redefines how and where redox reactions occur, shifting the reaction zone from the electrode surface to a dedicated chemical reactor. In doing so, the conventional electrochemical process is converted into a chemical process between two species

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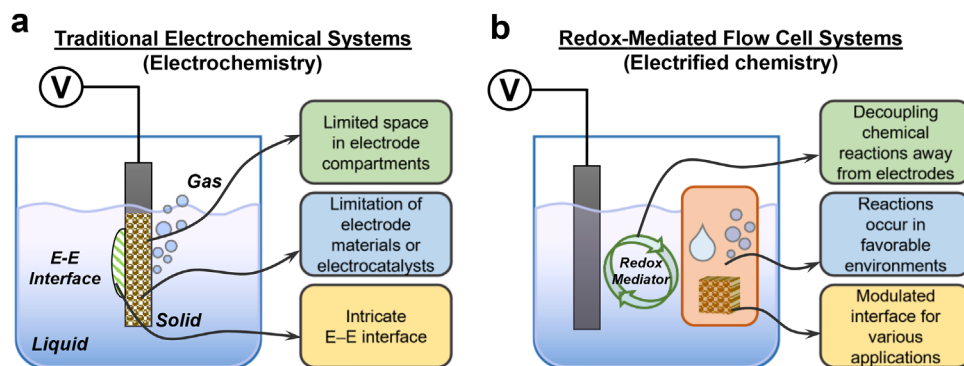


Figure 1. a,b) Illustrations of the half-cell of (a) a traditional electrochemical system and (b) a redox-mediated flow cell system, highlighting the limitations of the former and the advantages of the latter.

(phases), allowing reactions to take place in more favorable environments that alleviate the constraints of electrode geometry or conductivity. The resulting “electrified chemical process” enables a wide array of energy conversion, storage, and chemical production strategies. Moreover, the reaction interfaces in these systems are not limited to traditional liquid-solid contacts but can also include liquid-gas, solid-gas, or liquid-liquid interactions, providing extraordinary flexibility to accommodate diverse chemistries tailored to different applications.

Recent advancements have demonstrated the great potential and importance of redox-mediated flow cell systems. For instance, through judicious materials design, the capacity of organic redox electrolytes can be feasibly enhanced by enabling reactions with polymers derived from organic compounds with analogous core structures. This is evidenced by the observation of reversible redox reactions between RM and solid capacity boosters, along with a solid utilization rate that exceeds 85%.^[13–16] In addition, the precise alignment of redox potential through the implementation of doping methods for inorganic materials has been shown to yield an electrolyte capacity density that surpasses 110 Ah L⁻¹.^[17] Consequently, the objective of high-energy-density flow batteries, which seek to circumvent solubility constraints, has been successfully realized. Concurrently, molecular engineering techniques can be utilized to design RMs and systems for electrochemically induced aqueous CO₂ capture processes^[18–21] or to employ widely studied stable RMs for the recycling or recovery of lithium-ion battery (LIB) materials. Notwithstanding these advances, considerable challenges persist. A more profound comprehension of material requirements is imperative for achieving higher-performance systems, while lower costs and resolving engineering issues are indispensable steps in the scaling-up process. The employment of system simulation and modeling has been demonstrated to facilitate the resolution of numerous challenges without the necessity for experimental investigation.^[22] The integration of sophisticated tools, such as artificial intelligence, is paramount for the execution of high-throughput material screening.^[23,24]

Despite the diversity of their applications, redox-mediated flow cells are unified by a fundamental operational principle: the Electrochemical-Chemical (EC) cycle driven by redox potential differences between the mediator and the target material. In these systems, a redox-active mediator undergoes electrochem-

ical oxidation or reduction at an electrode, then diffuses or flows to a separate reaction site — typically a reactor tank — where it chemically exchanges electrons with a solid, liquid, or gaseous material for different applications. The mediator is then regenerated and circulated back to the surface of the electrode, forming a closed redox cycle that continuously transfers charges between the electrode and the reactive species.

The driving force behind this mediated redox reaction is the redox potential difference (ΔE) between the RM and the active material it targets. This potential difference may arise from three main sources: 1) Intrinsic potential mismatch between the mediator and the material, which thermodynamically promotes spontaneous charge transfer. 2) State-of-Charge (SOC)-dependent Nernstian potential variation, where the redox potential of the active material or mediator evolves during charge and discharge, enabling bidirectional reactions in reversible systems. 3) Environmental factors such as pH, temperature, or ionic strength, which modulate the redox potentials through Nernstian effects, thereby shifting the equilibrium of redox reactions.

For instance, in Single-Molecule Redox Targeting (SMRT) systems, the RM is designed to have nearly identical redox potential to that of the solid active material.^[25] In this case, the redox reaction proceeds not due to a large potential mismatch, but a dynamic equilibrium shift driven by changes in activity of the species involved in the reaction during operation, allowing reversible energy storage with high efficiency and low voltage loss. As shown in **Figure 2a**, the Fermi level of solid materials (E_F) in the reactor tank is matched with the standard equilibrium potential of RM (E_{RM}°) in the cell. In contrast, two-mediator redox-targeting systems often use two mediators with redox potentials straddling that of the solid material, ensuring directional control of oxidation and reduction processes during charge and discharge. While different from energy storage, in power generation (redox-mediated fuel cell) systems, redox-mediated chemical synthesis, waste recycling systems, etc., there isn't such a requirement for bidirectional reactions. Since the potential of reactant in the targeted reaction does not have to be matched, and the potential difference between the two provides the reaction driving force, a change in the SOC of the RM brings about a variation in the reaction driving force, ΔE , as shown in **Figure 2b**, which can lead to different reaction rates. Some other applications include pH-swing-based CO₂ capture systems based on the

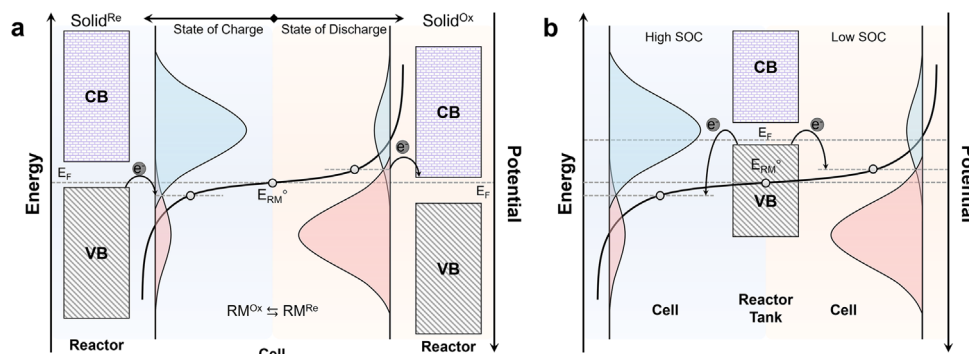


Figure 2. a) Working principle of the Single-Molecule Redox-Targeting (SMRT) reaction between an RM and a two-phase energy storage material at different states of charge/discharge in the positive electrode compartment. b) Change in response to the driving force ΔE due to varying SOC during the material oxidation process in the reactor tank.

Proton-Coupled Electron Transfer (PCET) process, which relies on the pH changes at different SOC for CO₂ capture (at high pH) and release (at low pH). Since the driving force is the chemical potential changes induced by pH variation, the redox potential requirement is generally relaxed.

With a good understanding of these common principles, thermodynamic driving forces, kinetic constraints, phase/interface versatility, etc., we can better appreciate how the same redox-mediated framework can be adapted for a wide variety of energy and chemical applications. In the following section, we explore how these fundamental mechanisms are implemented in specific systems and how they inform the selection and design of redox-active materials for distinct functionalities.

2. Energy Applications Enabled by Redox-Mediated Systems

2.1. Battery Research

Redox-mediated flow cell systems have been employed most extensively in the field of battery research, particularly with regard to redox flow batteries, including capacity boosting for high-energy-density redox flow batteries, redox-mediated strategy for specific problems such as dead material rejuvenation, and redox-mediated catalysis, etc.

2.1.1. Redox Targeting-Based Flow Batteries

Redox Targeting-based Flow Batteries (RT-FBs) represent an innovative technology in electrochemical energy storage by addressing the key limitations of conventional systems. The concept of redox-mediated energy storage systems has usually been called redox targeting (RT) systems, which was first introduced by Wang et al. in 2006,^[26] where soluble RMs were employed to transfer electrons between an electrode and spatially separated solid active materials. This process obviates the need for direct physical contact between the electrode and storage material, allowing the latter to be stored in dedicated reactor tanks rather than being electrically integrated into the current collector. As a result, a reversible redox-mediated process is established, forming the basis for what is now known as the RT-FBs.

In conventional flow batteries, the volumetric capacity is limited by the solubility of redox-active species.^[27–31] In contrast, by employing redox-targeting reactions, RT-FBs utilize solid capacity boosters housed in external electrolyte tanks, thereby overcoming solubility constraints and enabling higher energy density. In 2013, the RT strategy catalyzed the development of the first RT-FB architecture.^[32] In this system, high-capacity solid storage materials, including conventional battery materials such as LiFePO₄ or various redox-active solids, are placed in external tanks while the RMs circulate between these storage tanks and the electrode compartments, mediating charge transfer via an electrochemical-chemical cycle. This approach thus disruptively breaks the boundary between solid-phase energy storage of solid electrode batteries and liquid-phase energy storage of conventional redox flow batteries, combining the high energy density of the former with the scalability and operation flexibility of the latter.^[33–35]

In the early development of RT-FBs, dual RMs were used to bracket the redox potential of the solid storage material, ensuring efficient charge and discharge processes. However, this dual-mediator approach introduced voltage losses due to the inherent potential difference between the mediators.^[36] In addition, multiple redox species also increase the complexity of the electrolyte. To overcome this drawback, as shown in **Figure 3a**, the SMRT concept was then developed.^[25] In SMRT systems, a single RM with a redox potential closely matched to that of the solid material drives a reversible reaction dictated by the Nernstian potential difference. Today, RT-FBs continue to evolve, expanding into both aqueous^[37–40] and nonaqueous^[13,14,41–43] platforms and incorporating a diverse range of organic^[15,16,44] and inorganic RMs.^[40,42] The future success of RT-FBs depends critically on the precise design and assessment of both the RMs and the solid active materials, with particular focus on optimizing the reaction kinetics, stability, and overall compatibility.

2.1.2. Redox-Mediated Dead Material Rejuvenation and Catalysis

In addition to capacity boosting, the redox-mediated approach has also been employed to address critical issues in flow batteries. One example is the imbalanced capacity or electrolyte composition induced by side reactions upon operation, such as “dead

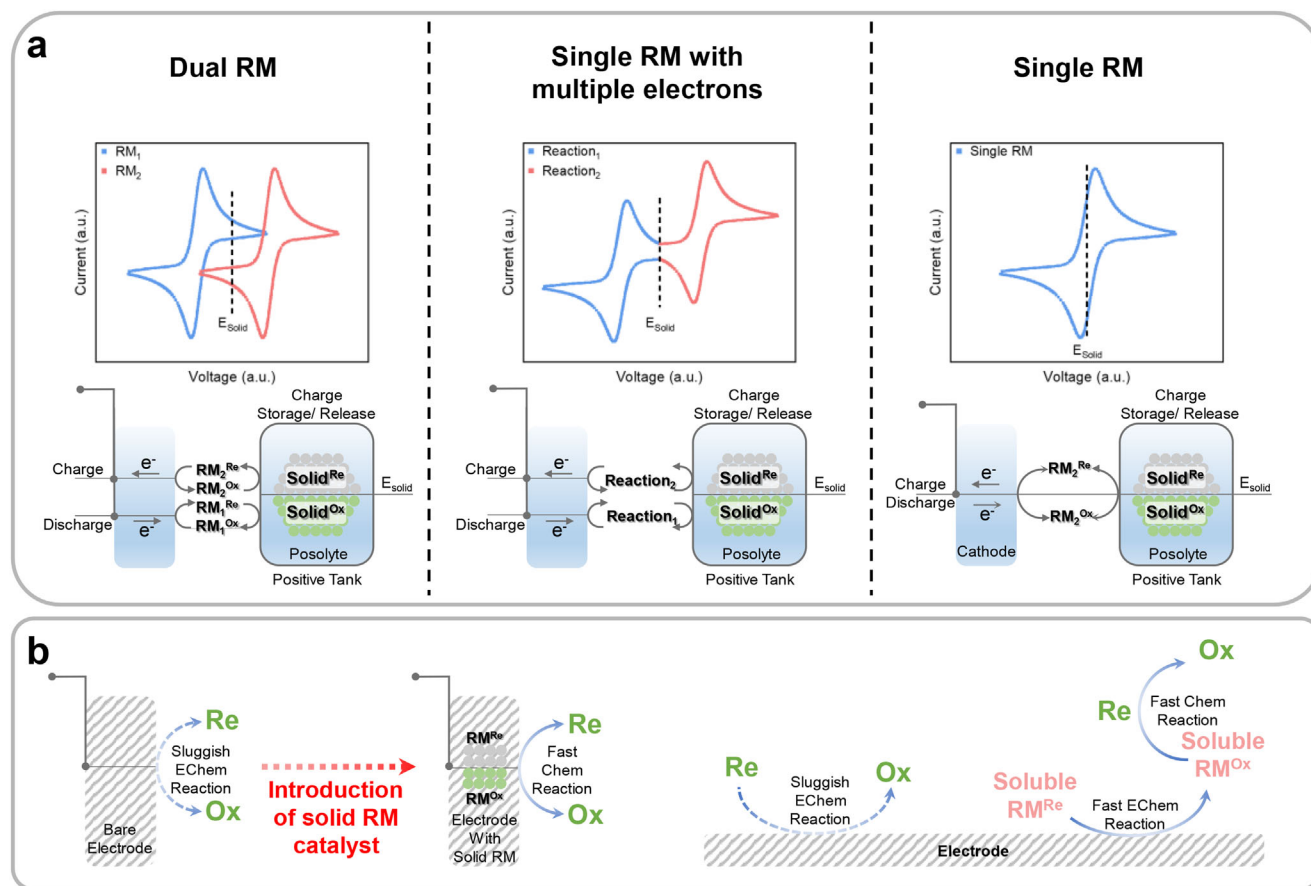


Figure 3. a) CV curves and reaction principle (with posolyte as the example) during charge/discharge of RT-FBs with dual RM, single RM with multiple electrons, and single RM. b) Solid RM (coated on the electrode) and soluble RM (dissolving in the electrolyte) catalysts in RFBs for enhanced reaction kinetics. Re/Ox: reduced/oxidized state of the redox-active substances in the electrolyte.

metal” generated by nonuniform plating/stripping of metal electrode or undesired HER reaction in metal-based flow battery systems. Metal-based batteries are widely studied due to their high capacity, while different from the homogeneous reaction of liquid electrolyte, repeated and inhomogeneous plating/stripping of metal may result in dead metal species electrically segregated from the electrode during the charge/discharge process, leading to irreversible capacity decay. A redox-mediated strategy elegantly solves the problem by directional electron transfer between RM and dead metal chemically, which restores their activity. Fast chemical reaction between 7,8-dihydroxyphenazine-2-sulfonic acid (DHPS) and zinc was used to improve the cyclability of zinc-based redox flow batteries.^[45] Besides dead metal reactivation, this approach has also been used to rejuvenate other dead solid battery materials, such as MnO₂.^[46–48] In addition, a redox-mediated OER reaction introduced to the catholyte in an alkaline Zn-[Fe(CN)₆]^{3-/4-} flow battery system could effectively rectify the imbalanced electrolyte composition induced by the HER side reaction on the Zn metal electrode.^[49] This strategy could also be applied to static cells,^[50,51] but the self-discharge caused by RM shuttling should be critically avoided.^[52]

With superior charge transfer kinetics and good selectivity between RM and the targeted substance, redox-mediated process can also be used to catalyze cell reactions either heterogeneously

or homogeneously (Figure 3b). Specifically, vanadium flow batteries (VFB), as the most mature RFB technology still face critical issues, such as limited energy efficiency and power density, partly due to the sluggish kinetics of the inner-sphere electron-transfer reaction of vanadium species on the electrodes. As an example, Prussian blue (PB) and a PB analogue (PBA) with identical redox potentials to VO²⁺/VO²⁺ and V^{2+/3+}, respectively, were grafted on the electrode.^[53,54] The reversible redox-targeting reactions between them facilitate the interfacial charge transfer and concomitantly inhibit hydrogen and oxygen evolutions. Moreover, based on the RT reactions between PB and [Fe(CN)₆]^{3-/4-}, PB and Br₂/Br⁻, which were first reported for capacity boosting in RT-FBs,^[38] PB-grafted electrodes were also reported to catalyze the sluggish redox reactions.^[55,56] The same strategy could also be implemented in the homogeneous process. For instance, polysulfide is a promising anolyte species for RFBs due to its low cost and high solubility in aqueous electrolyte, but the sluggish kinetics of polysulfide species, particularly in the reduction process, are still a challenge.^[57–61] A riboflavin sodium phosphate (FMN-Na) molecule was introduced in polysulfide electrolyte, as an active and durable molecular catalyst to transform the sluggish polysulfide reduction reactions (S₄²⁻ to S₂²⁻) into fast redox-mediated homogeneous catalytic processes,^[62] thus considerably enhancing the rate performance.

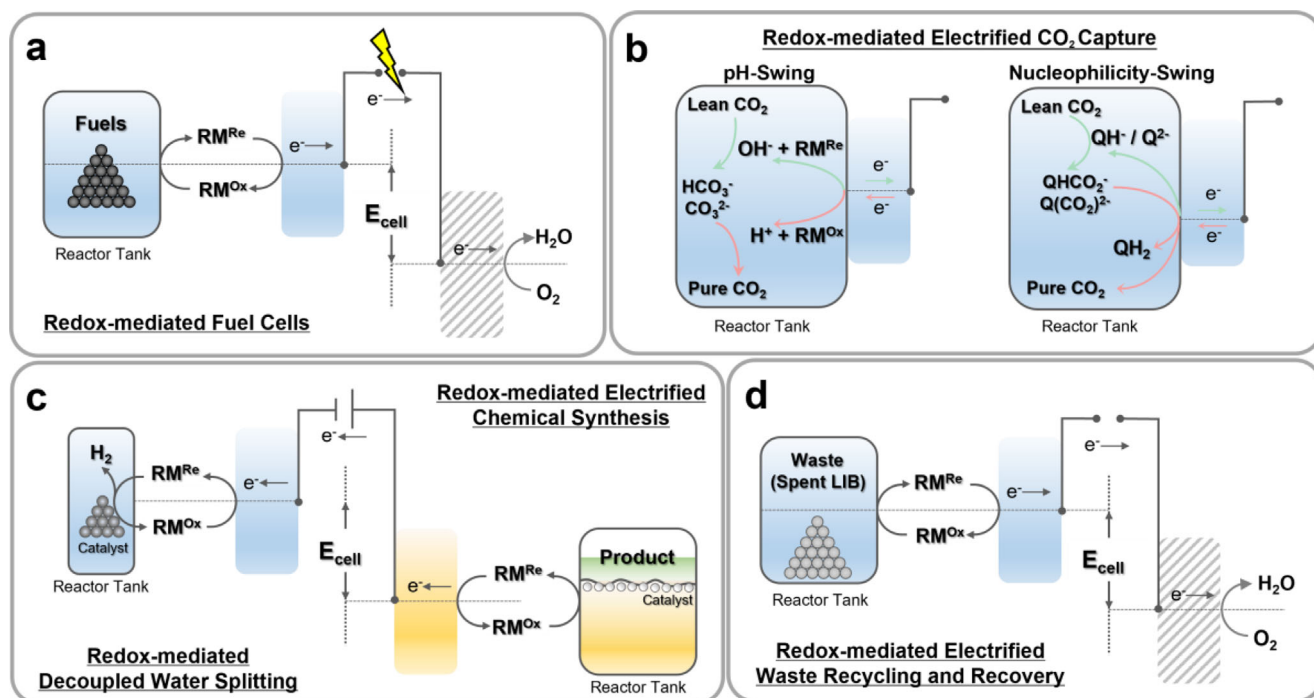


Figure 4. Working principle of a) Redox-mediated fuel cells (Solid fuels with negative redox potential paired with ORR as an example). b) Redox-mediated electrified CO₂ capture system with pH-swing and nucleophilicity-swing mechanisms. c) Redox-mediated electrified chemical synthesis (Decoupled HER and biphasic oxidation as an example). d) Redox-mediated electrified waste recycling and recovery (Recycling of spent LIB material paired with ORR as an example).

2.2. Redox-Mediated Fuel Cells

Redox-mediated fuel cells offer significant advantages by decoupling the reactions of fuels or oxygen from the intricate E-E interfaces, particularly suitable for power generation with feasible refueling and enhanced energy density. This approach leverages tailored RMs to shuttle electrons between electrodes and separate reactors, thereby enabling faster interfacial charge transfer and mass transport, and reducing limitations imposed by solubility and electrode surface area. One example is the aprotic lithium-air fuel cell, in which redox-mediated oxygen reduction reaction (ORR) proceeds in a separate gas diffusion tank, obviating the formation of insulating and insoluble discharge product Li₂O₂ on the electrode surface, thus preventing electrode passivation and pore clogging. Another example is the zinc-air fuel cell, where the zinc metal stored in the fuel tank is oxidized by an aqueous electrolyte-borne DHPS rather than on the electrode, thus showing enhanced operation flexibility and zinc utilization. Meanwhile, the integration of redox-mediated ORR reaction by using cobalt triisopropanolamine (CoTiPA) as RM is expected to eliminate issues of catalyst degradation associated with the heterogeneous catalysis.^[63] Similarly, in iron-air fuel cells, anthraquinone-2,6-disulfonate (2,6-AQDS) and 1,1'-bis(sulfonatopropyl)-4,4'-bipyridinium ((SPR)₂V) with a higher redox potential than iron metal were employed as RM. Iron metal stored in the fuel tank is chemically oxidized to Fe²⁺ in near-acidic conditions by RM, which eliminates the passivation problem of iron metal and improves the reaction kinetics. After Fe²⁺ ions cross the membrane, they are oxidized by oxygen via a

self-mediated ORR reaction by Fe^{2+/3+} in the polysolite. The pH changes induced by ORR lead to the precipitation of Fe³⁺ ions in the form of FeOOH in the polysolite tank.^[64] In hydrogen fuel cells,^[65] a H₄SiW₁₂O₄₀/H₅SiW₁₂O₄₀-based redox-mediated process allows hydrogen oxidation to occur under mild conditions, decreasing the reliance on high catalyst loading and improving overall system safety and flexibility.^[66] The integration of redox-mediated strategies paves the way for robust, scalable, and versatile power generation platforms, as demonstrated across several recent studies^[67–71] that emphasize improved energy conversion efficiency, reduced voltage losses, and enhanced durability of mediator-driven systems.

2.3. Redox-Mediated Electrified CO₂ Capture

Redox-active species can be utilized to mediate CO₂ capture and release in both protic and aprotic media (Figure 4b). In aprotic media, a redox-active species is first electrochemically reduced on the cathode and becomes nucleophilic. Subsequently, the reduced mediator chemically reacts with CO₂ via nucleophilic addition, forming an organic adduct to complete carbon capture. The adduct is then electrochemically oxidized on the anode to yield the initial mediator, which is no longer nucleophilic, thus releasing CO₂. It is the nucleophilicity swing that mediates CO₂ capture–release cycles.^[72,73] While in protic media, RMs reversibly undergo PCET reactions, affording a pH swing.^[21,74,75] Specifically, a mediator is electrochemically reduced on the cathode, raising the pH of the electrolyte. Subsequently, the alkaline

electrolyte reacts with CO_2 via a chemical acid-base reaction, capturing CO_2 in the form of inorganic carbonates via $\text{OH}^- + \text{CO}_2 \rightarrow \text{HCO}_3^-$ and $2\text{OH}^- + \text{CO}_2 \rightarrow \text{CO}_3^{2-} + \text{H}_2\text{O}$. The reduced mediator is then electrochemically oxidized on the anode, accompanied by the formation of protons, acidifying the electrolyte and reversing the acid-base reaction via $\text{H}^+ + \text{HCO}_3^- \rightarrow \text{H}_2\text{O} + \text{CO}_2$ and $2\text{H}^+ + \text{CO}_3^{2-} \rightarrow \text{H}_2\text{O} + \text{CO}_2$, and releasing CO_2 . When quinones are used as mediators in aqueous media, because the acid dissociation constants (pK_a) of hydroquinones are less than the pK_a (14) of water, the reduced quinones usually occur in three formats, including hydroquinone (QH_2), half deprotonated hydroquinone (QH^-), and fully deprotonated hydroquinone (Q^{2-}). In this scenario, besides the partial PCET-induced CO_2 capture, the deprotonated hydroquinones (QH^- , Q^{2-}) can directly react with CO_2 and form organic carbonates [QHCO_2^- , $\text{Q}(\text{CO}_2)_2^{2-}$] via nucleophilic addition. Thus, both pH- and nucleophilicity-swing mechanisms co-play a role in mediating CO_2 capture and release.^[18] When the pK_a values of mediators such as water-soluble phenazine/dihydrophenazine derivatives are higher than 14, pH-swing will become the sole mechanism of mediating carbon capture and release.^[19,76]

2.4. Redox-Mediated Electrified Chemical Synthesis

Redox-mediated strategies in electrosynthesis have recently attracted considerable attention for their ability to overcome limitations in conventional water splitting^[77–79] and other chemical production reactions^[80–85] (Figure 4c). In these systems, RMs facilitate decoupling of the electrochemical and chemical steps, allowing rigorous tailoring of the reaction environment, such as specific potential, temporal and spatial controls of each reaction step, etc. For instance, in conjunction with heterogeneous catalysts, an acidic $\text{V}^{2+/3+}$ and $\text{Ce}^{3+/4+}$ -based redox-flow system was reported for redox-mediated oxygen evolution reaction (OER) and hydrogen evolution reaction (HER), with which spatially decoupled water splitting was achieved.^[86] The same system has also been reported for hydrogen generation and desulfurization.^[87] Similarly, polyoxometalate was reported for both redox-mediated hydron production and hydrogen storage.^[88–90] In addition to the acidic conditions, an alkaline system with DHPS and $[\text{Fe}(\text{CN})_6]^{3-/4-}$ as the RMs for HER and OER reactions, respectively, has also been demonstrated with a low operation voltage and high faradaic efficiency for spatially decoupled water splitting.^[91] Based on $[\text{Fe}(\text{CN})_6]^{3-/4-}$ -mediated OER reaction, redox-mediated seawater splitting was also reported with controlled chloride oxidation.^[92] Moreover, redox-mediated strategy has also been used to electrified ammonia synthesis, in which a polyoxometalate redox species electrochemically reduced on the electrode was used in acidic media to chemically mediate the activation of molecular nitrogen on Fe-doped TiO_2 catalyst for hydrazine and ammonia synthesis.^[80]

Another redox-mediated electrosynthesis of 1,3,2-dioxathiolane 2,2-dioxide (DTD) by potassium permanganate (KMnO_4) was also reported with a reaction yield of 85% and faradaic efficiency of 72%. However, different from the gas product, which could be collected directly, DTD is a water-soluble chemical that is mixed with the redox species and needs further separation after the synthesis process.^[84] We have recently

demonstrated a redox-mediated electrified benzyl aldehyde synthesis based on a VFB cell. With benzyl alcohol as precursor and VO_2^+ as RM, it showed a near 100% selectivity without the formation of benzyl acid, and the benzyl aldehyde product could be collected directly since it is immiscible with the acidic electrolyte solution.

Besides aqueous media, the application of redox-mediated electrified chemical synthesis in immiscible aqueous/nonaqueous media could expand the scope of reactions with easy separation. An example is the 2,6-AQDS-mediated approach for H_2O_2 synthesis, using different cations in aqueous and organic phases. The AQDS anions shuttle between the two phases with protons and act as a charge carrier for the 2e^- ORR process.^[85] Another redox-mediated H_2O_2 synthesis was reported recently, for which, instead of using a single charge carrier, two RMs dissolved in the aqueous and organic phases, respectively, undergo PCET reaction at the water-oil interface and realize continuous operation.^[93] This kind of redox-mediated system with multiphase charge/ion transfer has implications for the electrified synthesis of a wide variety of chemicals.^[81]

In contrast to the homogeneous redox catalysis^[82,83] and heterogeneous electrocatalysis, the redox-mediated electrified chemical synthesis places significant emphasis on electron transfer between distinct phases. This approach provides a pathway to decouple the electrochemical reaction and the catalytic chemical reaction, and thus has greater flexibility for various chemical syntheses. The experience in redox catalysis could provide some useful insights into the future exploration of redox-mediated electrified chemical synthesis systems.

2.5. Redox-Mediated Electrified Waste Recycling and Recovery

Redox-mediated electrified materials recycling and recovery leverage the core principles of the electrochemical-chemical cycle to enable sustainable recovery of valuable species from spent materials, such as battery electrodes and other wastes (Figure 4d). Conventional recycling methods, including those based on high-temperature pyrometallurgy and chemical-intensive hydrometallurgy, often require significant energy input and harsh chemicals, leading to CO_2 emissions and secondary pollution. In contrast, redox-mediated electrified approaches operate upon regenerative reactions of RM with spent materials that selectively extract valuable species under ambient conditions.

In these systems, RMs are designed with redox potentials that enable fast reaction with spent materials. For instance, $[\text{Fe}(\text{CN})_6]^{3-/4-}$ was reported as the RM to break down spent LiFePO_4 ,^[94] a widely used LIBs cathode material by oxidation, during which Li^+ ions are extracted from the spent material when it's oxidized by $[\text{Fe}(\text{CN})_6]^{3-}$. The formed $[\text{Fe}(\text{CN})_6]^{4-}$ is regenerated on the electrode for a new round of reaction. Meanwhile, the extracted Li^+ ions are accumulated in the electrolyte and migrate to the counter electrode compartment, where lithium hydroxide (LiOH) is formed as the product. An ORR or HER^[95] process is paired on the counter electrode to complete the reaction. Recently, other RMs have been reported^[96–98] for such an electrified recycling process. In addition to recycling, redox-mediated electrified materials recovery has also been reported to restore the composition of spent materials in a scalable manner. One

example is the LiFePO_4 , which loses a significant amount of Li when it reaches the end of life. Using AQDS as RM, Li_xFePO_4 ($0 < x < 1$) black mass loaded in the reactor tank is reduced in the presence of Li^+ in the electrolyte, which accompanies Li^+ ions insertion, restoring Li composition. LiOH aqueous solution is added to the counter electrode compartment, where the OER or HOR process takes place to complete the full cell reaction.

Moreover, a redox-mediated electrochemical liquid-liquid extraction separation platform has been demonstrated for continuous separation and recycling of precious metals, specifically platinum and gold.^[99] Another system using an ion-selective potassium nickel hexacyanoferrate (KNiHCF) solid electrode as RM, which spontaneously oxidizes organic matter and uptakes ammonium ions (NH_4^+) and potassium ions (K^+) in manure wastewater and shows a nutrient selectivity of $\sim 100\%$.^[100,101] The redox-mediated flow design facilitates an electrified continuous operation without the need for external chemical reagents, thereby enabling the selective recovery of precious metals from complex multicomponent streams with high efficiency and sustainability. We anticipate that the same strategy could also be applied to the recycling of cement, plastic wastes, printed circuit board wastes, etc.

2.6. Other Emerging Applications

The redox-mediated concept has also been extended to other emerging applications, such as Thermally Regenerated Electrochemical Cycle (TREC)-based thermoelectric conversion systems.^[102–104] By pairing with $\text{Ni}_{0.8}\text{Co}_{0.2}(\text{OH})_2$ material stored in separate storage tanks, $[\text{Fe}(\text{CN})_6]^{3-/4-}$ electrolyte has achieved an absolute thermoelectric efficiency of 3.61% between 25 and 55 °C through redox-targeting reactions, more than double that of the conventional static TREC cells.^[105] The incorporation of solid material with high capacitive charge storage amplifies the entropy changes per cycle, considerably boosting the figure of merit of the TREC process.

Redox-mediated photoelectrochemical (PEC) systems overcome the classical coupling of light absorption, charge separation, and catalytic reaction by introducing redox shuttles that ferry photogenerated electrons (or holes) between photoelectrodes and spatially separated catalyst beds. For example, lithium-mediated ammonia synthesis integrated with a PEC cell employs $\text{Li}_3\text{N}/\text{Li}/\text{Li}^+$ as RMs between a Si photocathode and a catalytic N_2 -reduction chamber, having achieved high faradaic efficiencies at low bias.^[106] Similarly, a $\text{Fe}^{2+/3+}$ mediated photoelectrochemical water splitting was achieved by using BiVO_4 as the photocatalyst in a separate reactor.^[107] These approaches decouple photon-driven charge generation from the catalytic steps, permitting independent optimization of light absorbers, RMs, and catalysts, thus unlocking high-efficiency solar-driven chemistry in flow cell systems.

In the context of various redox-mediated flow cell systems, some redox mediators (RMs) have been notable due to their exceptional redox reversibility, stability, and capacity to function across a broad pH range, rendering them suitable for a range of applications. From a materials design perspective, the utilization of these widely adopted RMs frequently accomplishes the principle of simplicity. For instance, ferro/ferricyanide is employed

for energy storage at pH-neutral and alkaline conditions,^[38,108] redox-mediated OER,^[91] and LIBs materials recycling.^[109] While anthraquinone derivatives with superior stability and low cost are considered for energy storage systems^[40,44,110] and air batteries at various pH levels,^[12,70] LIBs materials recovery,^[111] and CO_2 capture,^[18,20] among other applications.

3. Materials Requirements for Optimal Redox-Mediated Flow Cell Systems

Achieving high-performance redox-mediated flow systems hinges on the rational design and selection of both soluble RMs and solid active materials, as well as the integration of the two. These materials must meet a set of rigorous criteria to support rapid, efficient, and durable electrochemical-chemical cycles, while also being economically viable for large-scale deployment (Figure 5).

3.1. Redox Potential

A fundamental requirement for the redox-mediated flow cell systems is that the redox potentials of RM and the targeted material must be precisely correlated. The electrochemical potential or redox potential difference (ΔE) provides the thermodynamic driving force of the reactions. For reversible redox processes like those in rechargeable energy storage systems, a suitable ΔE in both directions is required to drive the reactions for charge and discharge. The situation for CO_2 capture based on pH-swing is slightly different, for which the difference in chemical potential or ΔG drives the capture and release of CO_2 at different pH. In contrast, for unidirectional oxidation or reduction, like those in redox-mediated electrosynthesis or fuel cells, ΔE in the right direction is required to facilitate the reaction. Note that while reaction kinetics are generally enhanced at large ΔE , a suitable ΔE is desired for good energy efficiency and reaction selectivity.

In practice, the redox potential is dependent on the operating environment, such as temperature, pH, and ionic strength, as determined by the Nernst equation, thus affecting ΔE . For example, RMs based on the PCET process exhibit pH-dependent redox potential of different slopes, which can be exploited to adjust the magnitude of the thermodynamic driving force of the reaction. In addition, the effect of temperature on ΔG should not be neglected, and adjusting the temperature can lead to a variation of driving force, for which temperature-induced redox potential changes can even be used for waste heat harvesting through a thermoelectrical cycle.^[102,103] Thus, redox potential adjustment in a system represents a multidimensional design challenge, requiring holistic consideration of the thermodynamic tunability, balanced kinetics of the electrochemical-chemical cycle, and impact of operational conditions, to optimize the trade-off between energy efficiency and reaction kinetics.

Specifically, taking the SMRT system as an example, the redox potential alignment between RMs and Solid ensures bidirectional reversible reactions during the oxidation and reduction processes between them, with the driving force (ΔE) stemming from the Nernstian potential difference that arises from SOC variations.^[25] A recent study on LFP and $[\text{Fe}(\text{CN})_6]^{3-/4-}$ effectively

Materials Requirements for Optimal Redox-Mediated Flow Cell Systems

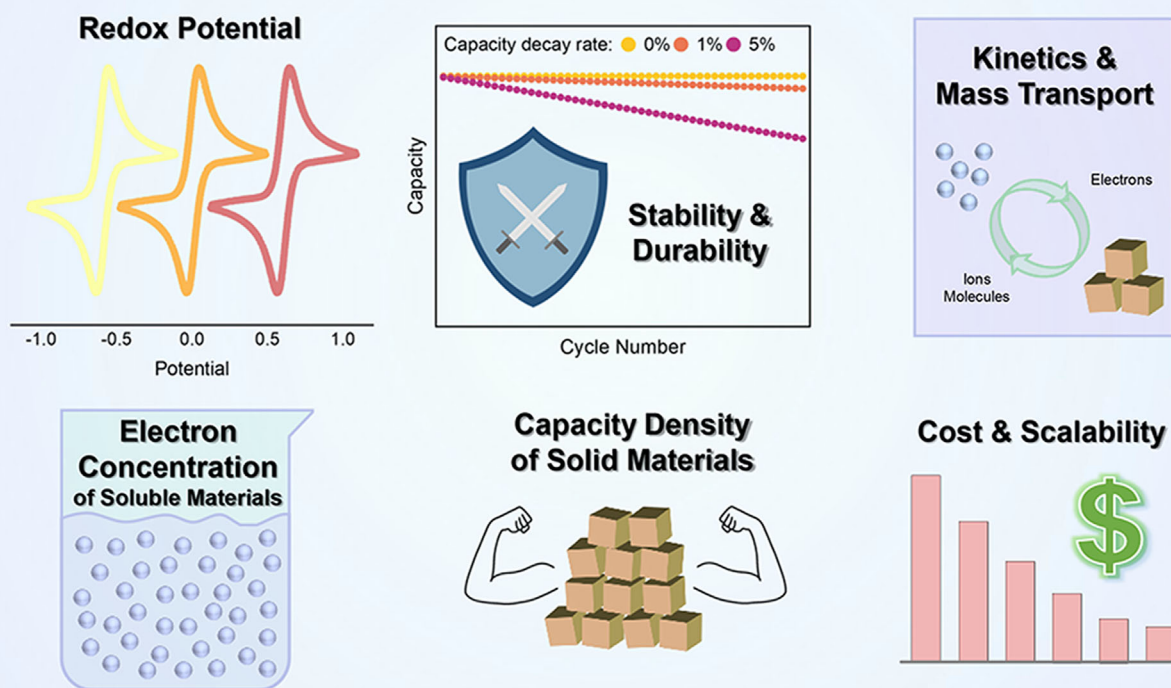


Figure 5. Materials requirements for optimal redox-mediated flow cell systems.

demonstrated the relationship between the degree of potential alignment and solid material utilization. The increase in potential of LFP caused by manganese doping directly affects the reaction capacity between the solid and the RM, highlighting the critical role of redox potential in the redox-mediated flow cell system.^[17] In addition, for redox-mediated electrified chemical synthesis involving sustained unidirectional reactions, such as DHPS-mediated HER, which exhibits an equilibrium potential ≈ 50 mV lower than HER, this provides the thermodynamic basis for the system's continuous operation.^[91]

3.2. Reaction Kinetics and Mass Transport

Kinetic characteristics of redox-mediated flow systems are determined by two coupled processes in tandem: the electrochemical reaction of RM at the surface of the electrode and the chemical reaction of RM with an active material in the reactor tank. Compared with conventional electrochemical cells, the redox-mediated process represents a spatially decoupled chemical step in addition to the electrochemical one, which requires balanced kinetics so that one process (generally that at the E-E interface) won't be limited by the other (chemical reaction in the tank). Equally important is mass transport bridging the two charge transfer events in the system, which requires well-designed flow fields, tailored electrode porosity, and low-viscosity electrolytes to maintain high diffusion coefficients and convective transport,

thereby minimizing concentration polarizations. Only by collectively optimizing electrode kinetics, reactor tank reaction rates, and mass transport pathways can a redox-mediated flow system achieve rapid dynamic responses and high power density.

For the electrochemical process in the electrode compartment, the RM undergoes an electron transfer reaction at the E-E interface. These are commonly described by Butler–Volmer formalism for overpotentials up to a few hundred millivolts before mass transport becomes limiting, or by Marcus–Hush–Chidsey theory when reorganization energies dominate the rate constant. In general, RMs with fast and reversible redox reactions at an appropriate potential are selected to react at the carbon felt electrode. Thus, enhancement of electrode kinetics involves: 1) Molecular engineering of RMs to optimize electronic coupling with the electrode and minimize steric hindrance for accelerated charge transfer rates. 2) Interfacial engineering by creating abundant catalytic sites for drastically enhanced reaction rate, particularly using high-surface-area electrodes or by surface treatment.

As for the redox-mediated reaction, the RM electrochemically oxidized or reduced at the electrode traverses into a separate reactor tank where it chemically reacts, often via heterogeneous electron transfer with an insoluble solid, a second liquid phase, a gas stream, or a supported catalyst. For optimal performance, this process is desired to proceed with both high rate and high selectivity for the desired redox reaction, which, however, relies on the nature of both RM and target material as the donor and acceptor of charges. Hence, the kinetics study of the redox-mediated

reaction should consider both phases, which is somewhat different from the classic electrode kinetics, where the density of charge in the electrode phase is a function of applied bias and generally not a limiting factor. In addition to the direct electron transfer with the active material, the redox-mediated step may also encompass RM adsorption/desorption equilibria and surface-mediated catalysis, typically following pseudo-first-order kinetics with respect to mediator concentration when the target material is in large excess.^[70,80,93] However, the detailed reaction mechanism must be determined on a case-by-case basis, as factors like surface composition, crystallinity (stacking of molecular materials), phase transition, and grain boundary can drastically alter the reaction pathways.

In addition, multiscale mass transport that bridges different reaction events plays a critical role in determining the overall system dynamics. While the mass transport in the electrode compartment is well studied, rapid ion transport through the membrane with tailored pore structures or ion-exchange properties is essential for reduced ohmic loss.^[112] In the reactor tank, concentration gradients of RM may build up around granular active material, leading to diffusion-limited regimes. Engineering solutions such as reducing granule size to shorten diffusion paths, increasing particle porosity to improve intra-particle transport, incorporating conducting additives to facilitate charge transfer, or inflating flow velocities to renew boundary layers, can shift the system into a kinetically controlled regime where the true chemical rate constant becomes accessible and tunable. Finally, materials design at the interface can further accelerate reaction rates. Surface functionalization, such as the introduction of redox-active surface groups or conductive coatings, can enhance electron-transfer rates and mediator adsorption, while maintaining facile desorption of products to prevent passivation. Combined with optimized reactor design, these strategies will ensure that the redox reaction is no longer the bottleneck, allowing redox-mediated systems to deliver rapid charge-discharge responsiveness and high power densities.

To balance the reaction rate of RM in the reactor tank with that in the electrode compartment, one flexibility of the redox-mediated flow system is that the loading of active materials in the tank could be feasibly altered, with which the reaction flux of RM could be adjusted to that of RM on the electrode, so that it won't limit the overall kinetics. Such a strategy has been proven to be effective in enhancing the utilization of capacity boosting materials and the rate performance of redox-mediated flow batteries.

3.3. Equivalent Electron Concentration of Redox Mediators

In addition to the general requirements, soluble RMs are the key components that shuttle charges between electrodes and active materials. Their physicochemical properties, most critically their solubility and electron-transfer characteristics, directly dictate the system's energy and power performances. Solubility is the foremost design metric for RMs, as it directly defines the concentration limit of redox active species and thus the achievable capacity (or energy) and reaction flux (or power) of the redox electrolytes. However, the solubility alone is not a sufficient measure for the above: the equivalent electron concentration and electrons transferred per redox cycle, by considering the materials

utilization, ultimately determine the volumetric capacity. High-concentration electrolytes, while boosting the capacity, often suffer from elevated viscosity, which hinders mass transport and increases pumping losses. To reconcile these competing operational demands, RMs involving multiple electron transfer are increasingly favored: single molecules capable of two or more electrons per redox event can match or exceed the effective electron concentration of monoelectronic species at lower total solute loading, thereby preserving fluidity and power performance.^[113] Moreover, high solubility and multielectron capacity synergize to minimize the concentration polarization under high current draw, enabling fast charge/discharge rates without sacrificing the voltage efficiency, thus achieving high power density.

3.4. Capacity Density of Solid Materials

Solid materials in redox-mediated flow systems (herein referring specifically to those used as capacity boosters, rather than the application-specific solids in waste recycling or fuel-cell reactors) store and supply charges via chemical reactions with RMs circulating through the externally housed reactor tanks. From a material design perspective, two interrelated criteria are paramount. First, the specific capacity directly determines the system's theoretical energy density: it is set by the specific gravimetric capacity (mAh/g) and, for flow systems, is more practically expressed as volumetric capacity (Ah/L). Designing high-capacity solids, therefore, demands materials with intrinsically large specific gravimetric capacity density and bulk density. Second, to maximize its capacity in practice, the solid material must be engineered to optimize the RM-solid reaction rate and ensure a higher utilization of the material. With sufficient loading of solid materials in the reactor tank, the key strategies include hierarchical porous structures combining high surface area with easy mediator access; particle-size engineering to minimize the diffusion length while forming large aggregates; surface functionalization using conducting additives or other coatings to improve its electronic conductivity and wettability for reduced activation barriers. By coupling high intrinsic capacity density with optimized microstructures and surface properties, solid capacity boosters can achieve exceptional volumetric and gravimetric performance and rapid kinetics in redox-mediated flow architectures.

3.5. Stability and Durability

In redox-mediated electrified systems, ensuring material stability and durability is paramount to achieving long-term operation. Compared to conventional systems, additional considerations should be made due to the coupled electrochemical and chemical processes. With the redox-mediated energy storage system as an example, RMs generally undergo more redox cycles than the conventional flow batteries, considering that for each charge/discharge cycle, the turnover time of RMs may be significantly larger than 1, depending on the ratio of RM and capacity booster. Thus, both the RM and solid materials must withstand repeated redox cycling without degradation. This includes chemical stability against side reactions, especially at harsh conditions, and mechanical stability for solid materials to prevent

structural failure. In contrast to conventional RFBs and analogous flow cell systems, RMs frequently function for extended durations at a constant SOC or within a predetermined SOC range during system operation. In some cases, this may lead to faster decay rates, such as the dimerization of anthraquinone derivatives in their reduced state under alkaline conditions. Conventional RFBs can circumvent this issue by constraining the SOC (this also limits the accessible capacity).^[114,115] However, for RT-FBs, constraining the SOC considerably impacts the driving force provided by the SOC, thereby imposing more stringent requirements on material stability. On the other hand, since the main capacity will be contributed by the solid capacity booster, a lower concentration is generally required in the electrolyte, which alleviates concentration-induced degradation, such as aggregation and precipitation.^[116–118] Designing RMs with a robust chemical backbone and selecting appropriate operating conditions can mitigate these potential issues. In addition to intrinsic structural stability, continuous flushing of electrolyte can potentially lead to damage to the structural integrity of solid materials. For reactions involving gases, the formation of gas bubbles on the substrate material may also cause mechanical damage to the system. One should note that many redox species, including those in the solid phase, despite being thermodynamically unstable, could still be employed for practical uses due to sufficiently sluggish side reactions. This principle underlies the operation of many commercial battery chemistries, from lead-acid batteries to state-of-the-art lithium-ion batteries. Such a kinetic constraint could also broaden the selection of viable redox systems for various applications.

3.6. Cost and Scalability

From an economic standpoint, the cost and scalability of redox-mediated systems are pivotal factors in determining their transition from laboratory prototypes to commercial solutions. This is especially the case for RT-FBs for large-scale energy storage. Materials should be composed of abundant elements and environmentally benign to ensure minimal raw material costs. The synthesis of RMs and fabrication of active solid materials must avoid complex, multistep processes that rely on expensive precursors or energy-intensive conditions. Instead, scalable manufacturing techniques such as solution processing, roll-to-roll printing, or other continuous fabrication methods are preferred. Additionally, the overall system design should strike a balance between performance enhancements and cost efficiency, ensuring that improvements in durability and kinetics do not incur unsustainable expenses. However, several scale-up challenges remain: optimizing packed-bed reactor design to minimize pressure drop while maintaining high material loadings, selecting suitable binders and conducting additives that preserve good wettability and electrical connectivity under high flow rates without compromising the cost, and developing simple, modular engineering solutions that facilitate plant-level integration. Therefore, prioritizing materials that are both cost-effective and compatible with large-scale production is essential for the widespread adoption of redox-mediated technologies, enabling them to be more viable than conventional energy storage and conversion systems.

4. Assessment Metrics and Technologies

With the key material requirements defined, rigorous evaluation frameworks that transform qualitative design goals into quantitative performance benchmarks are desired. In redox-mediated systems, the assessment must span from intrinsic material properties to full-cell behavior, integrating both experimental interrogation and computational insight.

4.1. Electrochemical Properties

For any redox-based systems, electrochemical properties are the first and most important metrics to be measured, regardless of the specific application. At the heart of any redox system lies its electrochemical signature. Cyclic voltammetry (CV) remains the most widely used workhorse for determining redox potentials, reversibility, and the number of electron transfers of both the soluble RMs and solid materials. Studies at different scan rates reveal whether the charge transfer is surface-controlled or diffusion-controlled and enable the extraction of the diffusion coefficient via the Randles–Ševčík relationship for the latter.^[119,120] Beyond CV, differential pulse voltammetry (DPV) with higher sensitivity in complex electrolytes (e.g., viscous or multicomponent flow battery solutions) by minimizing capacitive currents could also be used for redox potential determination.^[121] Moreover, electrochemical impedance spectroscopy (EIS) enables the deconvolution of distinct kinetic processes through their characteristic frequency responses (or time constant),^[122–124] while galvanostatic intermittent titration techniques (GITT) provide pseudo-steady-state diffusion coefficients and overpotential profiles under near-operando conditions.^[104,125]

4.2. Physical Properties

In a redox-mediated flow cell system, the conventional one-step electrochemical reaction is transformed into a spatially decoupled, two-step electrochemical-chemical process. This design imparts significant flexibility for real-time and *operando* spectroelectrochemical characterizations of both RMs and solid active materials using techniques, such as X-ray absorption spectroscopy (XAS),^[41,53] Ultraviolet–visible (UV–vis) spectroscopy,^[17,38] Fourier transform infrared spectroscopy (FTIR), etc.^[44,45] Depending on the technique, an optically transparent cell, compatible with the relevant detection signals, is connected to the outlet flow channel of a flow cell. Changes in properties such as concentration, bonding, valence state, and chemical environment of RMs in the electrolyte, as well as in the solid materials, are monitored. These observations help elucidate the underlying redox-target reaction mechanisms and provide a comprehensive picture of multiphase interactions at various stages of operation.

For capacity boosting materials in RT-FBs, the true volumetric capacity depends on the specific galvanometric capacity and packing density. The former can be calculated based on the number of electron transfers and molecular weight, while the packing density is generally determined through pellet compaction studies or water-displacement volume measurements. For crystalline

materials, the density value can also be obtained through the calculation of the unit cell volume.

Membrane selectivity for different ions, measured via diffusion cells with partitioned reservoirs, reveals crossover rates. Permeability, defined as the ratio of mediator-to-counterion transport numbers, guides membrane design for size and charge exclusion.^[126,127] Area-specific resistance (ASR) of membranes could be tested in similar cell configurations. While these metrics are crucial for overall flow-cell efficiency, they do not stem from the redox-mediated mechanism itself but rather reflect the universal trade-offs in membrane engineering. Low ASR is desired for high-power redox-mediated flow systems. Importantly, membrane crossover and fouling must be quantified to accurately interpret apparent capacity fade and to decouple true material degradation from membrane-induced losses during performance evaluation.

4.3. Computational Modeling and Simulations

Computational modeling becomes increasingly important for guiding materials and system design in redox-mediated flow architectures. Density functional theory (DFT) calculations have been widely employed to predict the standard potentials of redox molecules with different functional groups, varied solvation structures, and counter-ion environments, while other properties, such as solubility and stability, could also be predicted to guide the design and synthesis of these redox materials.^[128,129] In addition, molecular dynamics (MD) simulations have been employed to estimate the diffusion coefficients, electrolyte viscosity, and mediator aggregation tendencies.^[24,130] At the reactor level, finite element analysis (FEA) has been used to evaluate the pressure drops, shear profiles, and flow uniformity across serpentine, interdigitated, and porous flow-field geometries, informing the optimal distribution of electrolyte to minimize concentration polarization.^[131,132] Together, these multi-scale computational tools, accelerating materials screening with high-throughput calculations, also improve redox-mediated reactions and reactor designs, reducing experimental trial-and-error and expediting the realization of efficient and scalable redox-mediated flow systems. For enhanced throughput, especially for the complex electrochemical-chemical reaction cycles, machine learning is expected to play a more important role in providing useful guides to materials screening and system optimization.

4.4. Evaluations of the System Performance and Redox-Targeting Reactions

Most of the evaluations of the system performances are similar to the conventional redox flow systems or obey the principles of specific applications. Operational stability is assessed via long-term cycling protocols, whether constant-current (CC) or combined constant-voltage (CCCV), with capacity retention and fading rates reported per cycle or per calendar day. Different test protocols may lead to rather different stability conclusions. Compared with the capacity decay rate per cycle, the retention rate of the time scale is often more relevant to practical applications. Post-mortem analyses (e.g., high-performance liquid

chromatography (HPLC), nuclear magnetic resonance (NMR) of electrolytes, X-ray diffraction (XRD), and scanning electron microscope (SEM) of solids) reveal degradation pathways, such as molecule dimerization or solid-state phase changes, guiding materials improvements or regeneration strategies.^[115]

While conventional evaluation protocols for redox flow systems provide a useful reference, the most critical feature of redox-mediated flow systems is the redox-targeting reaction. Accurately assessing the depth and rate of this reaction is essential. For instance, in RT-FBs, the utilization of solid capacity-boosting materials should be quantified as the actual capacity increase relative to the theoretical capacity of the solid. This utilization, representing the depth of reaction, is closely tied to the reaction time. In some prior studies, high utilization rates have been reported, but these were often achieved under very low current densities, which may lead to misinterpretations of the material's true performance. To properly evaluate solid-state capacity boosters, the charge/discharge rate should be expressed in terms of specific current (C-rate or A/g), as is standard in studies of solid redox-active materials, rather than using areal current density (A/cm²), which is typical in flow battery systems. This distinction is important—increasing electrolyte volume or concentration, or increasing the mass loading of the capacity booster, while maintaining a constant areal current density, can enhance the utilization simply by extending the reaction time. Therefore, meaningful evaluation must consider both the intrinsic kinetics and the actual current per unit mass of the solid material.

Beyond the most intuitive indicators, such as capacity and coulombic efficiency in energy storage systems or faradaic efficiency in energy conversion systems, determining kinetic constants or reaction rates of the redox-targeting reactions often provides the most direct assessment of reactivity in redox-mediated flow systems. These measurements generally fall into two broad categories. The first involves electrochemical techniques, where reaction rates or reactivity are inferred from electrical signals. For example, the double-layer experiment, commonly used in redox-mediated studies,^[25] verifies the occurrence of the redox-targeting reaction by observing the depletion of reactants in the diffusion-limited region of CV or linear sweep voltammetry (LSV) tests. Similarly, the rotating disk electrode (RDE) coupled with open-circuit potential (OCP) measurements is also widely employed.^[45] Based on the Nernst equation, changes in OCP reflect variations in RM concentrations and the electrolyte's SOC, allowing the corresponding change in electrical parameters to be used to calculate the reaction rates. The second approach bypasses electrochemical measurements and instead tracks changes in the materials involved in the redox-targeting reaction, such as redox state interconversion or irreversible chemical transformations, using material characterization techniques. For instance, UV-vis spectroscopy can be employed at various stages of the reaction to directly quantify RM concentration and SOC, providing a non-electrochemical route to evaluating reaction kinetics.

5. Strategies for Designing Better Redox Materials

Here, we propose the design pathways and strategies for the development of redox-active materials, as shown in **Figure 6**, that can lead to enhanced redox-mediated flow systems.

Strategies for Designing Better Redox Materials

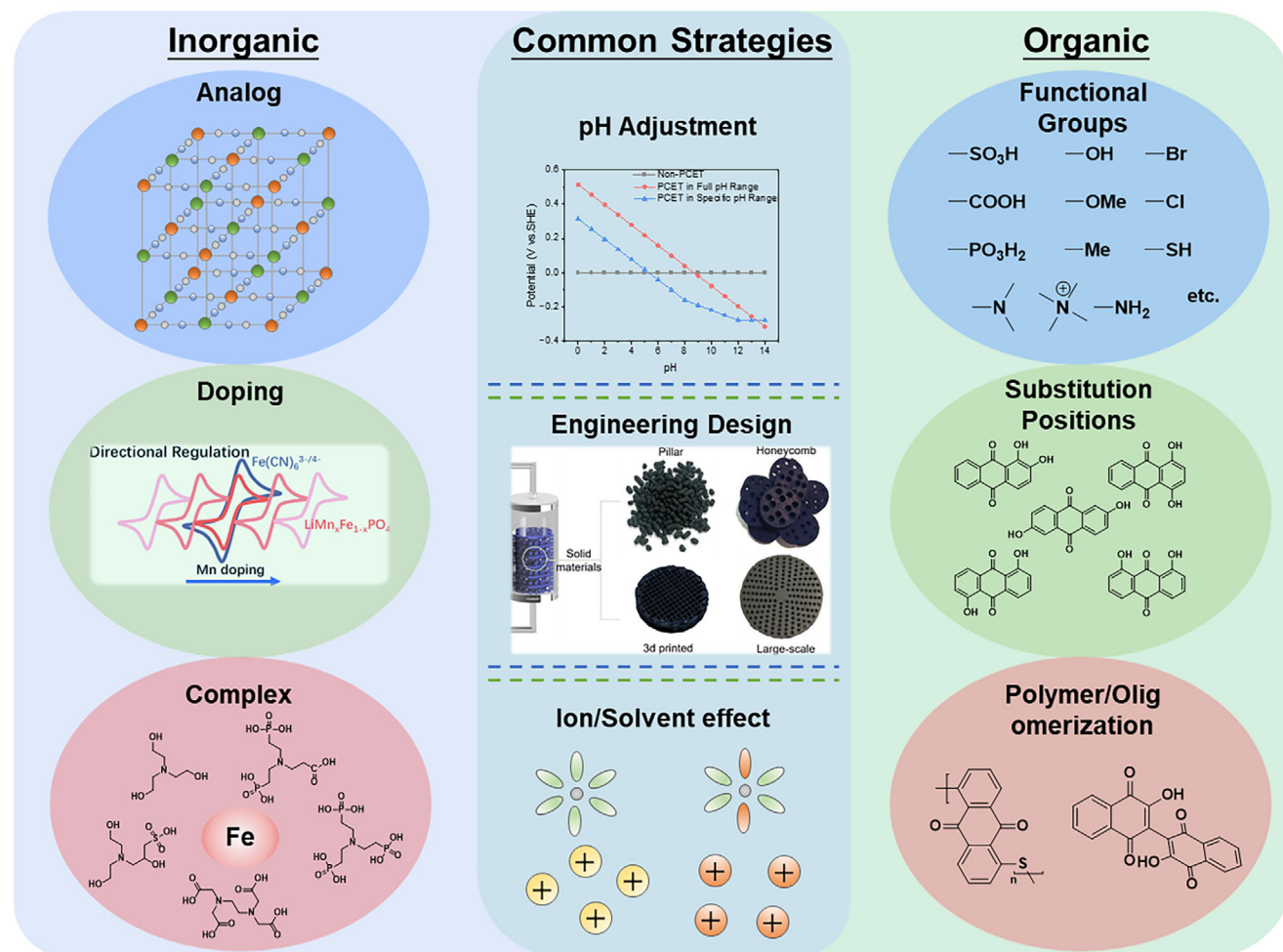


Figure 6. Strategies for designing better materials with examples adopted from literature.

5.1. Common Strategies

Adjusting the pH and controlling solvation are powerful strategies for tailoring the electrolyte composition to modulate the redox potentials and enhance the solubility of RMs. Considering RMs in the PCET process normally exhibit pH-dependent formal potentials, if two substances with similar potential have different sensitivities to pH (slope in Pourbaix diagram), then the driving force ΔE of the reaction can be controlled by adjusting the pH, and even oxidation or reduction at a specified pH can be achieved. Moreover, the solvation structure affects the electrochemical behavior of redox-active molecules. For instance, organic additives could change the redox potential of $[\text{Fe}(\text{CN})_6]^{3-/4-}$ in aqueous systems.^[37,133] In addition, the addition of ionic co-solutes (e.g., NH_4^+ , Mg^{2+} , SO_4^{2-}) stabilizes vanadium solvates via preferential bonding, widening the operating temperature window by 180% and raising the energy density by >30% in VFBS.^[134] Solvation control thus provides a route to fine-tune the ΔE without altering the core redox chemistry.

Ionic effect presents another strategy of interest for altering the physicochemical properties of redox species (e.g., solubility). For instance, mixed acid systems can substantially increase the concentration of vanadium ions in VFBS and thus achieve higher energy densities. The same strategy has been used in $[\text{Fe}(\text{CN})_6]^{3-/4-}$ electrolyte, where NH_4^+ ^[135] and Li^+ ions^[136] as counter ions can drastically enhance electrolyte concentrations to 2 M or even higher due to the low solvation energy. Furthermore, when Na^+ and K^+ are used together as mixed cations, solubility can be enhanced as well, which could be extended to other inorganic salts. The same principle applies to organic molecules such as AQDS and 1,8-dihydroxy-9,10-anthraquinone-2,7-disulphonic acid (DHAQDS). AQDS diammonium salt ($\text{AQDS}(\text{NH}_4)_2$), as an anolyte species for pH-neutral AORFBs, has a solubility of 1.9 M in water,^[137] which is more than 3 times that of the sodium salt. The solubility of DHAQDS is also governed by ion effects and the pKa of its sulfonate group.^[138,139] In acidic and Na^+ rich conditions, the sodium salt precipitates due to the common ion effect and charge shielding. Upon H^+ exchange, the protonated R-

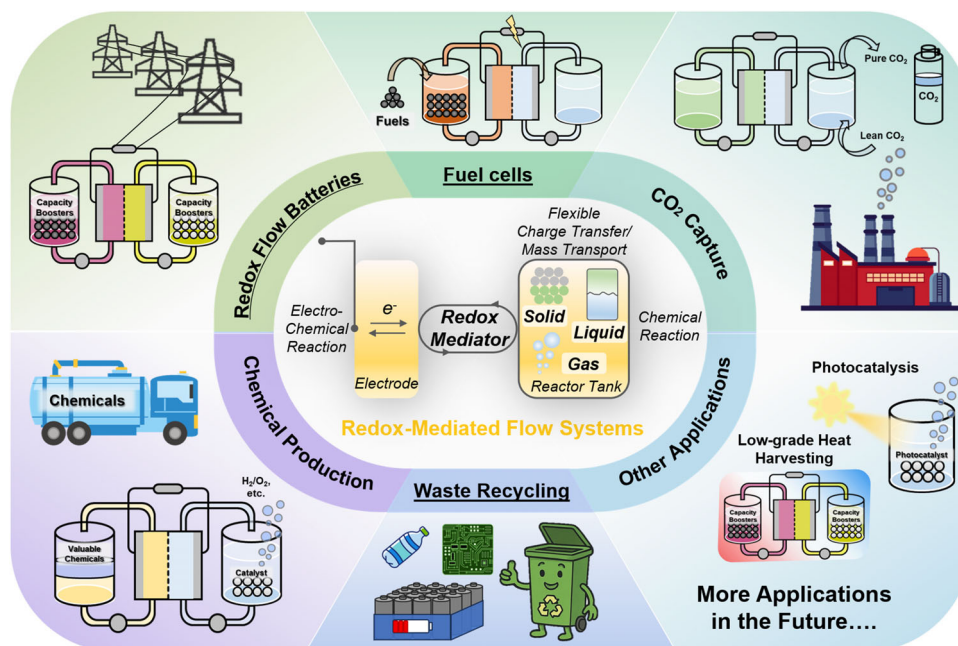


Figure 7. Various applications enabled by redox-mediated electrified flow cell systems.

SO_3H becomes soluble as a result of reduced ionic strength and enhanced interactions with water via hydrogen bonding. In addition to solubility, the cation or anion of a specific redox species could also lead to a variation of its redox potential. For instance, the redox potential of PB varies in NH_4^+ , Li^+ , Na^+ , and K^+ containing electrolytes.^[140]

Moreover, in the design of solid materials for RT systems, conducting additives are often added to improve conductivity, as well as binders to prevent the solid structure from being washed away and blocking the flow system.^[141] The selection of conducting additives and binders should not be random, as different choices affect the specific surface area and wettability, which can lead to a big difference in material utilization. In addition, at the laboratory level as well as when scaling up, the size of solid particles also affects the mass transport distance, and the design of the flow field will also have a nontrivial impact on the utilization of the material.

Beyond the routine materials development, recent advances in artificial intelligence and machine learning (AI-ML) have further accelerated discovery by rapidly screening vast chemical spaces against key performance criteria, especially for organic molecules.^[23] By training models on existing datasets of redox potentials, solubilities, and stability metrics, AI-ML workflows can predict promising core-substituent combinations with minimal human bias. When coupled with high-throughput virtual screening and DFT-calibrated surrogate models, these techniques narrow the candidate lists from thousands to the top few percent for experimental validation, reducing development timelines from year to month.^[24] The incorporation of AI-ML into the materials design process facilitates the identification of materials that align precisely with specific requirements for redox potential, reaction kinetics, and durability. This innovation is believed to transform the field of redox-mediated flow technologies, paving the way for new advancements in industry.

5.2. Design Strategies for Inorganic Redox Materials

The inorganic redox materials, particularly those used in soluble form, are rather limited compared to the organic counterparts. In addition to the common strategies mentioned above, one strategy for expanding the inorganic materials design is to form complexes with different ligands. With the same metal center, the properties of the complexes can be vastly different by changing the ligands. For example, in a metal complex based on $\text{Fe}^{2+/3+}$ centers in an aqueous solution, the redox potential can range from values close to OER for Fe-dcbpy/CN,^[142] $[\text{Fe}(\text{CN})_6]^{3-/4-}$ (in alkaline conditions)^[49,91] to values close to HER for Fe-TEA,^[143] Fe-TiPA,^[144] Fe-DIPSO, etc., or values in between for Fe-EDTA,^[145] Fe-DTPA,^[146] Fe-NTMPA,^[142] etc.^[147] In addition to homoleptic complexes, a heteroleptic strategy with mixed ligands provides greater flexibility in tailoring the physicochemical properties of complexes. Polyoxometalates consisting of multiple transition metal oxyanions through shared oxygen atoms represent another example of inorganic redox species providing diversified structures and redox properties.^[119,120] For inorganic solid materials, on the other hand, the discovery of analogs and the use of doping to alter electrochemical properties are commonly used. The roles of analogs and doping are presented here using two capacity boosting materials as examples. The first one is the difference in properties between PBA and PB. Changing the cation composition from Fe^{2+} to VO^{2+} dramatically increases the RT reactivity of PBA with the VFB catholyte from 10% utilization to over 60%.^[39] Additionally, doping can alter the material's electrochemical properties, improving its role in the RT system. For instance, the potential of LFP could be tuned by doping Mn, with which the resulting $\text{LiFe}_x\text{Mn}_{1-x}\text{PO}_4$ could match with $[\text{Fe}(\text{CN})_6]^{3-/4-}$ in neutral electrolyte and serve as a capacity booster.^[17]

5.3. Design Strategies for Organic Redox Materials

The design of organic materials for redox-mediated flow cells generally commences with the selection of a redox center, for which the standard potential should align with the desired operational window. Carbonyl-based redox centers (e.g., quinones),^[148–150] nitrogen-containing heterocycles (e.g., phenazines, viologens),^[151–154] and sulfur- or phosphorus-based systems^[155,156] each span distinct voltage ranges at standard conditions. Once a redox center is selected, the redox potential and kinetics of the species can be tuned by appending electron-donating groups (e.g., alkoxy, amino) to shift potentials more negative or electron-withdrawing moieties (e.g., sulfonate, nitrile, fluorinated aryl, halogen) to shift them more positive. The strategic placement of these substituents enables stepwise adjustments on the order of tens to hundreds of millivolts, facilitating precise matching between solid capacity boosters and soluble RMs.

To maximize electron capacity per molecule without compromising reaction dynamics, it is effective to incorporate multiple redox sites within each molecule or polymer repeating unit. Dimerization or oligomerization of the core units has been shown to double or triple the electron storage per entity.^[113,157,158] Conjugated linkages are effective in maintaining a low reorganization energy for facilitated electron-transfer kinetics. The rigidity of molecular backbones and the planar conjugation pathways have been demonstrated to promote outer-sphere electron transfers, thereby ensuring both the soluble RMs and redox polymers exhibit high-rate capability under flow conditions. In addition, it is imperative to exercise control over solubility. Hydrophilic substituents, including ionic groups such as $-\text{SO}_3^-$, $-\text{COO}^-$, and quaternary ammonium, have been shown to impart high aqueous solubility. This, in turn, enables mediator concentrations in the molar range with manageable viscosity. Conversely, the replacement of these with hydrophobic alkyl chains or the introduction of polymerizable vinyl/acetylene groups converts the same redox center into an insoluble form, thereby creating a static capacity booster. Such a solubility switch enables a single molecular scaffold to function in two distinct capacities: as a fluidic redox shuttle and a solid redox reservoir, respectively, without compromising the fundamental redox chemistry.

For long-term chemical and mechanical stability, steric protection and network formation mechanisms could be implemented. The synthetic approach is guided by considerations of cost, scalability, and sustainability. One-pot condensations, electrosynthesis, polymerizations, or mild catalytic routes are able to avoid the use of harsh reagents and minimize energy input.^[159–161] The employment of iterative computational-experimental workflows has been demonstrated to facilitate the identification of promising RM-solid pairs. This is achieved through DFT calculations to predict potential shifts and machine-learning models to forecast solubility and stability. Data-driven modeling or machine-learning approaches are expected to assist the search and screening of viable redox materials for redox-mediated flow cell systems in the future.

The integration of these strategies, including redox center selection, functional-group tuning, multisite incorporation, solubility modulation, steric and network stabilization, and green synthesis, would result in a unified design framework, facilitat-

ing customized fabrication of redox active materials that meet the conflicting requirements of redox potential alignment, high electron capacity, rapid kinetics, tunable solubility, and long-term durability, thereby establishing the foundation for advanced redox-mediated flow cell systems.

6. Summary and Outlook

Redox-mediated flow cell systems represent a paradigm shift for electrochemical technologies by decoupling energy storage, conversion, and chemical processing from the constraints of the traditional electrode-electrolyte interface. These systems unlock unprecedented flexibility, as shown in **Figure 7**, enabling applications ranging from high-energy-density flow batteries and fuel cells to sustainable CO_2 capture, chemical synthesis, and waste recycling. By leveraging soluble RMs and spatially separated reactive phases, redox-mediated architectures overcome limitations such as electrode passivation, sluggish kinetics, and low energy density inherent to conventional systems. Key advancements in materials design have significantly improved system efficiency, scalability, and durability, such as precise redox potential alignment, multielectron transfer mediators, and high energy density solids, etc.

However, challenges remain. Materials stability upon prolonged operation, crossover of redox species, cost of critical materials including redox species and membrane, reactor designs, and system integration are critical barriers, as those encountered by the conventional flow cells. Additionally, optimizing the interplay between electrochemical kinetics and chemical reaction rates in the decoupled systems requires a deeper mechanistic understanding. Addressing these challenges demands a holistic approach, integrating innovations in molecular/material engineering, reactor design, and advanced characterization techniques.

We believe the immediate research priorities should center on advancing the foundational and applied aspects of redox-mediated flow cell systems: 1) Establishing robust assessment metrics. This is critical to standardize the evaluation of system performance, particularly under realistic operating conditions. Key parameters include the utilization rates of solid materials, reaction depth, and energy efficiency, which require harmonized protocols to enable cross-system comparisons. 2) Developing advanced *in-situ* and *operando* characterization techniques such as synchrotron-based imaging and spectroscopy. It will provide real-time insights into the multiphase processes, degradation mechanisms, and phase transitions at different time and length scales. These tools are indispensable for unraveling the kinetic bottlenecks and optimizing the reaction pathways. 3) Accelerating computational material discovery through machine learning and multiscale modeling. This will promote rational design of tailored RMs and solids. Predictive frameworks for redox potential, solubility, and stability will reduce experimental trial-and-error, expediting the development of high-performance materials. 4) Integrating the redox-mediated technologies with renewable energy infrastructure (e.g., pairing flow batteries with solar/wind farms) or industrial processes (e.g., integrating CO_2 capture with emission-heavy sectors). Concurrent efforts are anticipated to foster interdisciplinary collaborations among chemists, engineers, and environmental scientists to jointly optimize materials, reactor designs, and system architecture.

The advancement of redox-mediated electrified systems has the potential to transform energy and chemical landscapes, providing scalable and sustainable solutions to the pressing global challenges. The decoupling of electrochemical and chemical processes has great potential to renovate the domains of energy storage, conversion, and utilization, thereby facilitating the seamless integration of these systems with renewable energy infrastructure. The inherent flexibility of these systems allows for adaptation to diverse applications, ranging from grid-scale long-duration energy storage to industrial decarbonization, while concurrently minimizing reliance on resource-intensive or environmentally harmful practices. Ultimately, redox-mediated electrified systems are poised to redefine the boundaries of electrochemical engineering. By aligning sustainability with performance, they will contribute to a resilient and electrified future, one in which energy storage, chemical synthesis, and environmental stewardship converge to address the pressing demands of climate change and sustainable development.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

energy storage and conversion, flow cell systems, materials design and assessment, redox-mediated, redox-targeting

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- [1] M. Park, J. Ryu, W. Wang, J. Cho, *Nat. Rev. Mater.* **2016**, 2, <https://doi.org/10.1038/natrevmats.2016.80>.
- [2] H. D. Yoo, E. Markevich, G. Salitra, D. Sharon, D. Aurbach, *Mater. Today* **2014**, 17, 110.
- [3] C. Liu, F. Li, L. P. Ma, H. M. Cheng, *Adv. Mater.* **2010**, 22, E28.
- [4] C. Yan, R. Xu, Y. Xiao, J.-F. Ding, L. Xu, B.-Q. Li, J.-Q. Huang, *Adv. Funct. Mater.* **2020**, 30, 201909887.
- [5] X. Yu, A. Manthiram, *Energy Environ. Sci.* **2018**, 11, 527.
- [6] R. Shi, L. Shang, T. Zhang, *ACS Appl. Energy Mater.* **2021**, 4, 1045.
- [7] H. Jiang, R. Luo, Y. Li, W. Chen, *EcoMat* **2022**, 4, 12199.
- [8] Y. Wang, Y. Zou, L. Tao, Y. Wang, G. Huang, S. Du, S. Wang, *Nano Res.* **2019**, 12, 2055.
- [9] R. Yan, Q. Wang, *Adv. Mater.* **2018**, 30, 1802406.
- [10] X. Wang, M. Zhou, F. Zhang, H. Zhang, Q. Wang, *Curr. Opin. Electrochem.* **2021**, 29, 100743.
- [11] F. Zhang, M. Gao, S. Huang, H. Zhang, X. Wang, L. Liu, M. Han, Q. Wang, *Adv. Mater.* **2022**, 34, 2104562.
- [12] M. Gao, Z. Wang, D. G. Lek, Q. Wang, *Nano Res. Energy* **2023**, 2, 9120045.
- [13] G. Lee, C. M. Wong, C. S. Sevov, *ACS Energy Lett.* **2022**, 7, 3337.
- [14] C. M. Wong, C. S. Sevov, *ACS Energy Lett.* **2021**, 1271, <https://doi.org/10.1021/acseenergylett.1c00143>.
- [15] E. Schröter, C. Stolze, A. Saal, K. Schreyer, M. D. Hager, U. S. Schubert, *ACS Appl. Mater. Interfaces* **2022**, 14, 6638.
- [16] E. Schroter, C. Stolze, J. Meyer, M. D. Hager, U. S. Schubert, *ChemSusChem* **2023**, 16, 202300296.
- [17] Y. Cai, H. Zhang, T. Wang, S. Xi, Y. Song, S. Rong, J. Ma, Z. Han, C. T. J. Low, Q. Wang, Y. Ji, *Joule* **2024**, 9, 101768.
- [18] Y. Jing, K. Amini, D. Xi, S. Jin, A. M. Alfaraidi, E. F. Kerr, R. G. Gordon, M. J. Aziz, *ACS Energy Lett.* **2024**, 9, 3526.
- [19] S. Jin, M. Wu, Y. Jing, R. G. Gordon, M. J. Aziz, *Nat. Commun.* **2022**, 13, 2140.
- [20] K. Amini, T. Cochard, Y. Jing, J. D. Sosa, D. Xi, M. Alberts, M. S. Emanuel, E. F. Kerr, R. G. Gordon, M. J. Aziz, *Nat. Chem. Eng.* **2024**, 1, 774.
- [21] S. Pang, S. Jin, F. Yang, M. Alberts, L. Li, D. Xi, R. G. Gordon, P. Wang, M. J. Aziz, Y. Ji, *Nat. Energy* **2023**, 8, 1126.
- [22] N. J. Matteucci, C. T. Mallia, B. J. Neyhouse, F. R. Brushett, *J. Electrochem. Soc.* **2025**, 172, 070539.
- [23] P. Gao, D. Kochan, Y.-H. Tang, X. Yang, E. G. Saldanha, *J. Power Sources* **2025**, 629, 236035.
- [24] F. Wang, Z. Ma, J. Cheng, *J. Am. Chem. Soc.* **2024**, 146, 14566.
- [25] M. Zhou, Q. Huang, T. N. Pham Truong, J. Ghilane, Y. G. Zhu, C. Jia, R. Yan, L. Fan, H. Randriamahazaka, Q. Wang, *Chem* **2017**, 3, 1036.
- [26] Q. Wang, S. M. Zakeeruddin, D. Wang, I. Exnar, M. Grätzel, *Angew. Chem.* **2006**, 118, 8377.
- [27] D. G. Kwabi, Y. Ji, M. J. Aziz, *Chem. Rev.* **2020**, 120, 6467.
- [28] L. Zhang, R. Feng, W. Wang, G. Yu, *Nat. Rev. Chem.* **2022**, 6, 524.
- [29] Z. Li, Y. C. Lu, *Adv. Mater.* **2020**, 32, 2002132.
- [30] Z. Yuan, Y. Yin, C. Xie, H. Zhang, Y. Yao, X. Li, *Adv. Mater.* **2019**, 31, 1902025.
- [31] X. Wei, W. Xu, J. Huang, L. Zhang, E. Walter, C. Lawrence, M. Vijayakumar, W. A. Henderson, T. Liu, L. Cosimbescu, B. Li, V. Sprenkle, W. Wang, *Angew. Chem. Int. Ed. Engl.* **2015**, 54, 8684.
- [32] Q. Huang, H. Li, M. Gratzel, Q. Wang, *Phys. Chem. Chem. Phys.* **2013**, 15, 1793.
- [33] F. Pan, J. Yang, Q. Huang, X. Wang, H. Huang, Q. Wang, *Adv. Energy Mater.* **2014**, 4, 201400567.
- [34] J. R. Jennings, Q. Huang, Q. Wang, *J. Phys. Chem. C* **2015**, 119, 17522.
- [35] Q. Huang, J. Yang, C. B. Ng, C. Jia, Q. Wang, *Energy Environ. Sci.* **2016**, 9, 917.
- [36] Y. G. Zhu, Y. Du, C. Jia, M. Zhou, L. Fan, X. Wang, Q. Wang, *J. Am. Chem. Soc.* **2017**, 139, 6286.
- [37] J. Yu, L. Fan, R. Yan, M. Zhou, Q. Wang, *ACS Energy Lett.* **2018**, 3, 2314.
- [38] Y. Chen, M. Zhou, Y. Xia, X. Wang, Y. Liu, Y. Yao, H. Zhang, Y. Li, S. Lu, W. Qin, X. Wu, Q. Wang, *Joule* **2019**, 3, 2255.
- [39] Y. Cheng, X. Wang, S. Huang, W. Samarakoon, S. Xi, Y. Ji, H. Zhang, F. Zhang, Y. Du, Z. Feng, S. Adams, Q. Wang, *ACS Energy Lett.* **2019**, 4, 3028.
- [40] M. Zhou, Y. Chen, M. Salla, H. Zhang, X. Wang, S. R. Mothe, Q. Wang, *Angew. Chem. Int. Ed. Engl.* **2020**, 59, 14286.
- [41] M. Zhou, Y. Chen, Q. Zhang, S. Xi, J. Yu, Y. Du, Y.-S. Hu, Q. Wang, *Adv. Energy Mater.* **2019**, 9, 201901188.
- [42] Y. Zhou, G. Cong, H. Chen, N.-C. Lai, Y.-C. Lu, *ACS Energy Lett.* **2020**, 5, 1732.
- [43] E. C. Self, F. M. Delnick, R. E. Ruther, S. Allu, J. Nanda, *ACS Energy Lett.* **2019**, 4, 2593.
- [44] S. Huang, H. Zhang, M. Salla, J. Zhuang, Y. Zhi, X. Wang, Q. Wang, *Nat. Commun.* **2022**, 13, 4746.
- [45] S. Huang, Z. Yuan, M. Salla, X. Wang, H. Zhang, S. Huang, D. G. Lek, X. Li, Q. Wang, *Energy Environ. Sci.* **2023**, 16, 438.

- [46] J. Lei, Y. Yao, Z. Wang, Y.-C. Lu, *Energy Environ. Sci.* **2021**, *14*, 4418.
- [47] Q. Wang, W. Zhou, Y. Zhang, H. Jin, X. Li, T. Zhang, B. Wang, R. Zhao, J. Zhang, W. Li, Y. Qiao, C. Jia, D. Zhao, D. Chao, *Natl. Sci. Rev.* **2024**, *11*, nwae230.
- [48] J. Cao, K. Yu, J. Zhang, B. Lu, J. Yu, S. Huang, F. Zhang, *ACS Sustainable Chem. Eng.* **2024**, *12*, 6320.
- [49] S. Huang, M. Li, Y. Song, S. Xi, C. Wu, Z. W. J. Ang, Q. Wang, *Adv. Mater.* **2024**, *36*, 2406366.
- [50] X. Qian, T. Chen, Y. Wang, Q. Zhang, W. Li, J. Fu, *Angew. Chem. Int. Ed. Engl.* **2025**, *64*, 202412989.
- [51] C. Jin, T. Liu, O. Sheng, M. Li, T. Liu, Y. Yuan, J. Nai, Z. Ju, W. Zhang, Y. Liu, Y. Wang, Z. Lin, J. Lu, X. Tao, *Nat. Energy* **2021**, *6*, 378.
- [52] J. Chen, Z. Cheng, Y. Liao, L. Yuan, Z. Li, Y. Huang, *Adv. Energy Mater.* **2022**, *12*, 202201800.
- [53] F. Zhang, S. Huang, X. Wang, C. Jia, Y. Du, Q. Wang, *Nano Energy* **2018**, *52*, 292.
- [54] L. Xia, T. Long, W. Li, F. Zhong, M. Ding, Y. Long, Z. Xu, Y. Lei, Y. Guan, D. Yuan, Y. Zhang, C. Jia, L. Sun, Q. Sun, *Small* **2020**, *16*, 2003321.
- [55] Q. Zhang, H. Jiang, S. Liu, Q. Wang, J. Wang, Z. Zhou, K. Cai, Q. Lai, Q. Wang, *J. Power Sources* **2024**, *601*, 234286.
- [56] X. Jia, H. Kang, G. Hou, S. Lu, Y. Yao, Q. Wang, W. Qin, X. Wu, *ACS Catal.* **2023**, *13*, 3692.
- [57] J. Zhang, W. Zhou, D. Zhao, Y.-C. Lu, D. Chao, *Nat. Rev. Electr. Eng.* **2025**, *2*, 215.
- [58] D. Ma, B. Hu, W. Wu, X. Liu, J. Zai, C. Shu, T. Tadesse Tsega, L. Chen, X. Qian, T. L. Liu, *Nat. Commun.* **2019**, *10*, 3367.
- [59] M. Ding, H. Fu, X. Lou, M. He, B. Chen, Z. Han, S. Chu, B. Lu, G. Zhou, C. Jia, *ACS Nano* **2023**, *17*, 16252.
- [60] J. Liu, W. Zhou, R. Zhao, Z. Yang, W. Li, D. Chao, S.-Z. Qiao, D. Zhao, *J. Am. Chem. Soc.* **2021**, *143*, 15475.
- [61] M. Gao, S. Huang, F. Zhang, Y. M. Lee, S. Huang, Q. Wang, *Mater. Today Energy* **2020**, *18*, 100540.
- [62] J. Lei, Y. Zhang, Y. Yao, Y. Shi, K. L. Leung, J. Fan, Y.-C. Lu, *Nat. Energy* **2023**, *8*, 1355.
- [63] H. Zhang, S. Huang, M. Salla, J. Zhuang, M. Gao, D. G. Lek, H. Ye, Q. Wang, *ACS Energy Lett.* **2022**, *7*, 2565.
- [64] M. Gao, Y. Song, X. Zou, M. Salla, H. Zhang, Z. Wang, S. Huang, Q. Wang, *Adv. Energy Mater.* **2023**, *13*, 202301868.
- [65] C. W. Anson, S. S. Stahl, *Chem. Rev.* **2020**, *120*, 3749.
- [66] H. Liu, Y. Yin, X. Cao, H. Cheng, Y. Xie, C. Wu, *J. Am. Chem. Soc.* **2024**, *146*, 5274.
- [67] A. M. Fenton, Y. Ashraf Gandomi, C. T. Mallia, B. J. Neyhouse, M. A. Kpeglo, W. E. Exson, C. T.-C. Wan, F. R. Brushett, *ACS Omega* **2022**, *7*, 40540.
- [68] Y. Song, L. Xia, M. Salla, S. Xi, W. Fu, W. Wang, M. Gao, S. Huang, S. Huang, X. Wang, X. Yu, T. Niu, Y. Zhang, S. Wang, M. Han, M. Ni, Q. Wang, H. Zhang, *Angew. Chem. Int. Ed. Engl.* **2024**, *63*, 202314796.
- [69] C. T. Mallia, F. R. Brushett, *Chem Commun (Camb)* **2024**, *60*, 11363.
- [70] S. Huang, H. Zhang, J. Zhuang, M. Zhou, M. Gao, F. Zhang, Q. Wang, *Adv. Energy Mater.* **2022**, *12*, 202103622.
- [71] S. Huang, Q. Wang, *Acc. Mater. Res.* **2023**, *4*, 693.
- [72] B. Gurkan, F. Simeon, T. A. Hatton, *ACS Sustainable Chem. Eng.* **2015**, *3*, 1394.
- [73] Y. Liu, H. Z. Ye, K. M. Diederichsen, T. Van Voorhis, T. A. Hatton, *Nat. Commun.* **2020**, *11*, 2278.
- [74] H. Xie, W. Jiang, T. Liu, Y. Wu, Y. Wang, B. Chen, D. Niu, B. Liang, *Cell Rep. Phys. Sci.* **2020**, *1*, 100046.
- [75] S. Jin, M. Wu, R. G. Gordon, M. J. Aziz, D. G. Kwabi, *Energy Environ. Sci.* **2020**, *13*, 3706.
- [76] Z. Wei, S. Xu, J. Liu, K. Zhu, M. Tang, A. Li, X. Wang, K. P. Tan, Y. Li, Z. Meng, Y. Jing, *ChemRxiv* **2025**, <https://doi.org/10.26434/chemrxiv-2025-rmfmq>.
- [77] F. Zhang, Q. Wang, *ACS Mater. Lett.* **2021**, *3*, 641.
- [78] V. Muralidharan, S. Jayasubramanian, H.-W. Lee, *EES Catalysis* **2024**, *2*, 522.
- [79] F. Zhang, Q. Wang, *Curr. Opin. Electrochem.* **2022**, *35*, 101097.
- [80] X. Wang, J. Yang, M. Salla, S. Xi, Y. Yang, M. Li, F. Zhang, M.-K. Zhu, S. Huang, S. Huang, Y.-W. Zhang, Q. Wang, *Angew. Chem. Int. Ed. Engl.* **2021**, *60*, 18721.
- [81] A. Alfaraidi, J. Zhou, A. Ni, N. Su, D. Xi, J. Lee, E. Kwan, M. Aziz, R. Liu, *ChemRxiv* **2025**, <https://doi.org/10.26434/chemrxiv-2025-44rd7>.
- [82] R. Francke, R. D. Little, *Chem. Soc. Rev.* **2014**, *43*, 2492.
- [83] R. Francke, R. D. Little, *Curr. Opin. Electrochem.* **2023**, *40*, 101315.
- [84] C. Hu, Z. Zhao, W. Wang, W. Zou, S. Liu, X. Fang, X. Su, N. Zheng, *Joule* **2025**, *9*, 101807.
- [85] A. T. Murray, S. Voskian, M. Schreier, T. A. Hatton, Y. Surendranath, *Joule* **2019**, *3*, 2942.
- [86] V. Amstutz, K. E. Toghill, F. Powlesland, H. Vrubel, C. Comninellis, X. Hu, H. H. Girault, *Energy Environ. Sci.* **2014**, *7*, 2350.
- [87] P. Peljo, H. Vrubel, V. Amstutz, J. Pandard, J. Morgado, A. Santasalo-Aarnio, D. Lloyd, F. Gumy, C. R. Dennison, K. E. Toghill, H. H. Girault, *Green Chem.* **2016**, *18*, 1785.
- [88] M. D. Symes, L. Cronin, *Nat. Chem.* **2013**, *5*, 403.
- [89] B. Rausch, M. D. Symes, G. Chisholm, L. Cronin, *Science* **2014**, *345*, 1326.
- [90] J. J. Chen, M. D. Symes, L. Cronin, *Nat. Chem.* **2018**, *10*, 1042.
- [91] F. Zhang, H. Zhang, M. Salla, N. Qin, M. Gao, Y. Ji, S. Huang, S. Wu, R. Zhang, Z. Lu, Q. Wang, *J. Am. Chem. Soc.* **2021**, *143*, 223.
- [92] T. Liu, C. Lan, M. Tang, M. Li, Y. Xu, H. Yang, Q. Deng, W. Jiang, Z. Zhao, Y. Wu, H. Xie, *Nat. Commun.* **2024**, *15*, 8874.
- [93] D. Xi, Y. Wu, Y. Li, Z. Yan, R. Y. Liu, M. J. Aziz, *Nat. Chem.* **2025**, *1*, <https://doi.org/10.1038/s41557-025-01940-7>.
- [94] J. Yu, X. Wang, M. Zhou, Q. Wang, *Energy Environ. Sci.* **2019**, *12*, 2672.
- [95] X. Jia, H. Kang, G. Hou, W. Wu, S. Lu, Y. Li, Q. Wang, W. Qin, X. Wu, *Angew. Chem. Int. Ed. Engl.* **2024**, *63*, 202318248.
- [96] P. Zhu, J. Hu, J. Hu, Y. Yang, W. Sun, Y. Yang, G. Zou, H. Hou, X. Ji, *ACS Energy Lett.* **2024**, *9*, 569.
- [97] B. Zhang, L. He, J. Wang, Y. Liu, X. Xue, S. He, C. Zhang, Z. Zhao, L. Zhou, J. Wang, Z. L. Wang, *Energy Environ. Sci.* **2023**, *16*, 3873.
- [98] J. Liu, Z. Liu, Z. Xiao, Y. Zhu, J. Wang, G. Ji, Y. Liu, B. Sun, G. Zhou, *Adv. Mater.* **2025**, <https://doi.org/10.1002/adma.202503450>.
- [99] S. R. Cotty, A. Faniyan, J. Elbert, X. Su, *Nat. Chem. Eng.* **2024**, *1*, 281.
- [100] R. Wang, K. Yang, C. Wong, H. Aguirre-Villegas, R. Larson, F. Brushett, M. Qin, S. Jin, *Nat. Sustain.* **2023**, *7*, 179.
- [101] R. Wang, A. Moeller, H. Tang, P. F. Cobra, M. Qin, S. Jin, *ACS Energy Lett.* **2024**, *9*, 4673.
- [102] H. Zhang, Z. Wang, Q. Wang, *Appl. Phys. Lett.* **2023**, *122*, 146725.
- [103] H. Zhang, Q. Wang, *Chem. Phys. Rev.* **2021**, *2*, 0044616.
- [104] H. Zhang, Z. Wang, H. Li, M. Salla, Y. Song, S. Huang, S. Huang, X. Wang, K. Liu, G. Xu, J. Huang, C.-W. Qiu, Q. Wang, *Joule* **2023**, *7*, 1515.
- [105] H. Zhang, F. Zhang, J. Yu, M. Zhou, W. Luo, Y. M. Lee, M. Si, Q. Wang, *Adv. Mater.* **2021**, *33*, 2006234.
- [106] H. Huang, W. Tu, L. Fang, Y. Xiao, F. Niu, H. Zhu, X. Zhu, L. Wang, Y. Xiong, J. Feng, H. Li, M. Zhang, Y. Zhang, Y. Yao, Y. Zhou, H. Li, Z. Zou, *ACS Energy Lett.* **2023**, *8*, 4235.
- [107] Y. Zhao, C. Ding, J. Zhu, W. Qin, X. Tao, F. Fan, R. Li, C. Li, *Angew. Chem. Int. Ed. Engl.* **2020**, *59*, 9653.
- [108] T. Pérez, F. Zhang, M. Á. Muñoz, L. Lubian, S. Xi, R. Sanz, Q. Wang, J. Palma, E. Ventosa, *Adv. Energy Mater.* **2021**, *12*, 202102866.
- [109] J. Yu, M. Salla, H. Zhang, Y. Ji, F. Zhang, M. Zhou, Q. Wang, *Energy Storage Mater.* **2020**, *29*, 216.
- [110] J. Ma, Y. Cai, S. Rong, M. Zhang, J. Zhang, Z. Han, S. Liu, W. Yang, C. Zhang, F. Zhang, Y. Ji, *ACS Nano* **2025**, *27538*, <https://doi.org/10.1021/acsnano.5c06747>.

- [111] Z. Wang, S. Huang, X. Wang, M. Salla, H. Wang, Q. Wang, *ACS Energy Lett.* **2025**, *10*, 3064.
- [112] H. Wang, S. Y. Sayed, E. J. Lubner, B. C. Olsen, S. M. Shirurkar, S. Venkatakrishnan, U. M. Tefashe, A. K. Farquhar, E. S. Smotkin, R. L. McCreery, J. M. Buriak, *ACS Nano* **2020**, *14*, 2575.
- [113] X. Fang, Z. Li, Y. Zhao, D. Yue, L. Zhang, X. Wei, *ACS Mater. Lett.* **2022**, *4*, 277.
- [114] M.-A. Goulet, L. Tong, D. A. Pollack, D. P. Tabor, S. A. Odom, A. Aspuru-Guzik, E. E. Kwan, R. G. Gordon, M. J. Aziz, *J. Am. Chem. Soc.* **2019**, *141*, 8014.
- [115] Y. Jing, E. W. Zhao, M.-A. Goulet, M. Bahari, E. M. Fell, S. Jin, A. Davoodi, E. Jónsson, M. Wu, C. P. Grey, R. G. Gordon, M. J. Aziz, *Nat. Chem.* **2022**, *14*, 1103.
- [116] W. M. Carvalho, L. Cassayre, D. Quaranta, F. Chauvet, R. El-Hage, T. Tzedakis, B. Biscans, *J. Energy Chem.* **2021**, *61*, 436.
- [117] Z. Xiang, W. Li, K. Wan, Z. Fu, Z. Liang, *Angew. Chem. Int. Ed. Engl.* **2023**, *62*, 202214601.
- [118] Z. Xiang, T. Ren, M. Huang, W. Li, L. Wang, K. Wan, Z. Fu, Z. Liang, *Angew. Chem. Int. Ed. Engl.* **2025**, *64*, 202416184.
- [119] X. Wang, M. Gao, Y. M. Lee, M. Salla, F. Zhang, S. Huang, Q. Wang, *Nano Energy* **2022**, *93*, 106864.
- [120] F. Ai, Z. Wang, N.-C. Lai, Q. Zou, Z. Liang, Y.-C. Lu, *Nat. Energy* **2022**, *7*, 417.
- [121] B. Hu, H. Li, H. Fan, J. Song, *Energy Storage Mater.* **2023**, *59*, 102789.
- [122] M. Gao, M. Salla, F. Zhang, Y. Zhi, Q. Wang, *J. Power Sources* **2022**, *527*, 231180.
- [123] D. Xi, A. M. Alfaraidi, J. Gao, T. Cochard, L. C. I. Faria, Z. Yang, T. Y. George, T. Wang, R. G. Gordon, R. Y. Liu, M. J. Aziz, *Nat. Energy* **2024**, *9*, 479.
- [124] T. Wang, D. Xi, M. J. Aziz, *ACS Appl. Energy Mater.* **2024**, *7*, 6578.
- [125] H. Zhang, D. G. Lek, S. Huang, Y. M. Lee, Q. Wang, *Adv. Mater.* **2022**, *34*, 2202266.
- [126] T. Y. George, E. M. Fell, K. Lee, M. S. Emanuel, M. J. Aziz, *Energy Adv.* **2024**, *3*, 2910.
- [127] T. Y. George, I. C. Thomas, N. O. Haya, J. P. Deneen, C. Wang, M. J. Aziz, *ACS Appl. Mater. Interfaces* **2023**, <https://doi.org/10.1021/acsaami.3c14173>.
- [128] S. Rong, J. Ma, H. Zhang, J. Yu, T. Wang, Y. Cai, Z. Han, Y. Ji, *Energy Storage Mater.* **2024**, *69*, 103389.
- [129] N. Rahbani, P. de Silva, E. Baudrin, *ChemSusChem* **2023**, *16*, 202300482.
- [130] J. Asenjo-Pascual, I. Salmeron-Sanchez, P. Mauleón, M. Agirre, A. C. Lopes, O. Zugazua, E. Sánchez-Díez, J. R. Avilés-Moreno, P. Ocón, *J. Power Sources* **2023**, *564*, 232817.
- [131] B. Xiong, Y. Li, Y. Ding, J. Wang, Z. Wei, J. Zhao, X. Ai, J. Fang, *J. Power Sources* **2023**, *564*, 232814.
- [132] L. F. Pilonetto, F. Staciaki, E. Nóbrega, E. B. Carneiro-Neto, J. da Silva, E. Pereira, *Chem. Eng. J.* **2023**, *474*, 145336.
- [133] Z. Wang, H. Zhang, Q. Wang, *Nano Energy* **2025**, *134*, 110515.
- [134] V. Murugesan, Z. Nie, X. Zhang, P. Gao, Z. Zhu, Q. Huang, L. Yan, D. Reed, W. Wang, *Cell Rep. Phys. Sci.* **2021**, *2*, 100323.
- [135] J. Luo, B. Hu, C. Debruler, Y. Bi, Y. Zhao, B. Yuan, M. Hu, W. Wu, T. L. Liu, *Joule* **2019**, *3*, 149.
- [136] X. Li, Y. Yao, C. Liu, X. Jia, J. Jian, B. Guo, S. Lu, W. Qin, Q. Wang, X. Wu, *Angew. Chem. Int. Ed. Engl.* **2023**, *62*, 202304667.
- [137] B. Hu, J. Luo, M. Hu, B. Yuan, T. L. Liu, *Angew. Chem. Int. Ed. Engl.* **2019**, *58*, 16629.
- [138] M. R. Gerhardt, L. Tong, R. Gómez-Bombarelli, Q. Chen, M. P. Marshak, C. J. Galvin, A. Aspuru-Guzik, R. G. Gordon, M. J. Aziz, *Adv. Energy Mater.* **2016**, *7*, 201601488.
- [139] B. Huskinson, M. P. Marshak, C. Suh, S. Er, M. R. Gerhardt, C. J. Galvin, X. Chen, A. Aspuru-Guzik, R. G. Gordon, M. J. Aziz, *Nature* **2014**, *505*, 195.
- [140] S. Phadke, R. Mysyk, M. Anouti, *J. Energy Chem.* **2020**, *40*, 31.
- [141] N. J. Matteucci, C. T. Mallia, B. J. Neyhouse, M. V. Majji, F. R. Brushett, *Curr. Opin. Electrochem.* **2023**, *42*, 101380.
- [142] G. S. Nambafu, A. M. Hollas, S. Zhang, P. S. Rice, D. Boglaenko, J. L. Fulton, M. Li, Q. Huang, Y. Zhu, D. M. Reed, V. L. Sprenkle, G. Li, *Nat. Commun.* **2024**, *15*, 2566.
- [143] Y. Ji, F. Zhang, M. Zhou, J. Yu, Q. Wang, *Int. J. Hydrogen Energy* **2020**, *45*, 18888.
- [144] J. Yang, W. Wei, C. Zhou, H. Yan, H. Che, L. Hao, X. Tan, A. W. Robertson, T.-S. Wu, Y.-L. Soo, A. Tang, Z. Sun, *Angew. Chem. Int. Ed. Engl.* **2025**, *64*, 202414452.
- [145] A. S. N. M. T. Srivastava, *J. Power Sources* **1989**, *27*, 119.
- [146] S. E. Waters, B. H. Robb, M. P. Marshak, *ACS Energy Lett.* **2020**, *5*, 1758.
- [147] J. Yang, C. Zhou, Y. Xiang, Y. Bi, X. Tan, A. W. Robertson, Y. Cheng, Z. Sun, *ChemSusChem* **2025**, 2500384.
- [148] Z. Yang, L. Tong, D. P. Tabor, E. S. Beh, M.-A. Goulet, D. De Porcellinis, A. Aspuru-Guzik, R. G. Gordon, M. J. Aziz, *Adv. Energy Mater.* **2017**, 201702056.
- [149] F. Hasan, V. Mahanta, A. A. A. Abdelazeez, *Adv. Mater. Interfaces* **2023**, *10*, 202300268.
- [150] K. Lin, Q. Chen, M. R. Gerhardt, L. Tong, S. B. Kim, L. Eisenach, A. W. Valle, D. Hardee, R. G. Gordon, M. J. Aziz, M. P. Marshak, *Science* **2015**, *349*, 1529.
- [151] G. Tang, K. Peng, Y. Liu, J. Fang, W. Wu, Z. Yang, T. Xu, *Angew. Chem. Int. Ed. Engl.* **2025**, *64*, 202501458.
- [152] X.-L. Lv, P. Sullivan, H.-C. Fu, X. Hu, H. Liu, S. Jin, W. Li, D. Feng, *ACS Energy Lett.* **2022**, *7*, 2428.
- [153] C. Wang, X. Li, B. Yu, Y. Wang, Z. Yang, H. Wang, H. Lin, J. Ma, G. Li, Z. Jin, *ACS Energy Lett.* **2020**, *5*, 411.
- [154] A. Hollas, X. Wei, V. Murugesan, Z. Nie, B. Li, D. Reed, J. Liu, V. Sprenkle, W. Wang, *Nat. Energy* **2018**, *3*, 508.
- [155] Q. Chen, W. Wang, X. Li, W. Guo, Y. Fu, *Proc. Natl. Acad. Sci. U. S. A.* **2022**, *119*, 2202449119.
- [156] P. Sang, Q. Chen, D. Y. Wang, W. Guo, Y. Fu, *Chem. Rev.* **2023**, 1262.
- [157] C. Xie, C. Wang, Y. Xu, T. Li, Q. Fu, X. Li, *Nat. Energy* **2024**, *9*, 714.
- [158] Y. Yao, Z. Wang, Z. Li, Y. C. Lu, *Adv. Mater.* **2021**, *33*, 2008095.
- [159] Y. Jing, M. Wu, A. A. Wong, E. M. Fell, S. Jin, D. A. Pollack, E. F. Kerr, R. G. Gordon, M. J. Aziz, *Green Chem.* **2020**, *22*, 6084.
- [160] H. Ji, Z. Zhao, C. Zhang, X. Li, *Chem. Sci.* **2024**, *15*, 13185.
- [161] Z. Zhao, T. Li, C. Zhang, M. Zhang, S. Li, X. Li, *Nat. Sustain.* **2024**, *7*, 1273.